

The Fabius Function and Rvachev’s Up-Function

Exact arithmetic, probability, Fourier analysis, and corrected sharp asymptotics

Human-readable synthesis of the formal development

Vladimir Reshetnikov’s `ProveIt` repository

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Abstract

The Fabius function is the unique smooth distribution function that is constant outside $[0, 1]$, symmetric about $1/2$, and satisfies the self-similar differential equation $F'(x) = 2F(2x)$ on the left half of the unit interval. Folding it about the midpoint gives Rvachev’s compactly supported even function up, one of the canonical examples of a C^∞ function with compact support that is given by an exact refinement equation. The same object admits several sharply different descriptions: a contraction fixed point, a probability law of an infinite weighted sum of independent uniform random variables, an infinite convolution, an infinite product of sinc functions, a Thue–Morse translate sum, and a family of exact rational recurrences.

This article develops these descriptions and proves their equivalence. Particular emphasis is placed on two subjects for which the formal development goes substantially beyond the usual short accounts. First, every dyadic value is evaluated by an explicit finite Thue–Morse/moment formula, and also by a fast terminating recursion that removes one set bit of the numerator at each step. Second, the small-argument asymptotic is derived by an exact negative-Laplace decomposition and a quantitative saddle-point argument. The correct main term contains a genuine nonconstant periodic fluctuation in the lower Lambert coordinate. Its Fourier coefficients are explicit Gamma–zeta values. Omitting that fluctuation, as in a frequently reproduced elementary formula, destroys the claimed inverse-logarithmic error bound. The first saddle correction and the full all-orders expansion are also described.

The presentation follows the theorem structure of the Lean formalization under `Analysis/FabiusFunction`, while translating the machine-oriented decomposition into conventional mathematical arguments. Source-level endpoint and normalization defects are corrected rather than repeated. In particular, a centered finite-spline approximation is given with an explicit uniform error, and this approximation is turned into a primitive-recursive dyadic algorithm proving computability of the bounded Fabius function in the sense of Grzegorczyk.

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1 Historical note and scope

The probability distribution now attached to the name of Fabius is older than Fabius's two-page note. It occurs in the work of Jessen and Wintner on infinite convolutions, was rediscovered several times, and entered a different mathematical culture through the compactly supported “finite function” of V. A. Rvachev. Fabius used it in 1966 as a particularly economical probabilistic construction of a nowhere analytic C^∞ function [2]. Rvachev's normalization emphasizes the even compactly supported density, whereas the Fabius normalization emphasizes its cumulative distribution function. Arias de Reyna's 1982 paper gave a systematic treatment of the compactly supported function, including its Fourier product, probability interpretation, partitions of unity, derivatives, and dyadic values; an English translation appeared in 2017 [4]. His later arithmetic paper organized the exact rational sequences and denominator questions [5].

The present article is not merely a paraphrase of those papers. It follows the proof dependency graph of the Lean development in the `ProveIt` repository [8]. That development first constructs the bounded distribution function, then derives the Rvachev and signed-global forms, and finally builds the arithmetic and asymptotic layers on top. This order has two advantages. It prevents identities valid for the signed extension from being silently applied to the bounded distribution, and it makes the exact dyadic algorithms independent of any numerical approximation.

The formalization also settles several questions that are only implicit, conjectural, or incorrectly normalized in the source material. Most importantly, the sharp endpoint asymptotic contains a nonconstant periodic function of an exact lower-Lambert phase. The periodic term is not a cosmetic improvement: every nonzero Fourier mode is explicit and nonzero, and an elementary formula without it fails at the error scale usually claimed. The development further proves a full inverse-Lambert expansion, establishes an exact bit-recursive evaluator for every dyadic rational, and supplies an absolutely and uniformly convergent Fourier–Legendre expansion of up .

Our convention is fixed throughout:

- F is the bounded cumulative distribution function, equal to 0 on $(-\infty, 0]$ and to 1 on $[1, \infty)$;
- up is the even compactly supported density on $[-1, 1]$;
- $\mathcal{F}$ is the signed global extension satisfying $\mathcal{F}'(x) = 2\mathcal{F}(2x)$ on all of $\mathbb{R}$.

All logarithms are natural logarithms. We write $L = \log 2$ in the asymptotic sections, $\text{wt}_2(n)$ for the sum of the binary digits of n , and $\text{sinc } z = \sin z/z$ with the removable value 1 at the origin.

2 Three functions that must not be conflated

The literature often uses the same letter F for three related, but globally different, functions. That economy of notation is harmless in a local calculation and dangerous in a long proof. We therefore begin by separating them.

2.1 The bounded Fabius distribution function

Definition 2.1 (Bounded Fabius function). A function $F : \mathbb{R} \rightarrow [0, 1]$ is called a *bounded Fabius function* if

$$F(x) = 0 \quad (x \leq 0), \quad (1)$$

$$F(x) = 1 \quad (x \geq 1), \quad (2)$$

$$F(1-x) = 1 - F(x) \quad (0 \leq x \leq 1), \quad (3)$$

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}), \quad (4)$$

and F is infinitely differentiable on $\mathbb{R}$.

The derivative equation on the right half follows from symmetry:

$$F'(x) = 2F(2-2x), \quad \tfrac{1}{2} \leq x \leq 1. \quad (5)$$

In particular $F'(x) > 0$ for $0 < x < 1$, once positivity in the interior is known. Thus F is a genuine cumulative distribution function, not merely a smooth interpolation between 0 and 1.

Because F is constant on both exterior half-lines and is C^∞ , every derivative of positive order vanishes at both endpoints:

$$F^{(m)}(0) = F^{(m)}(1) = 0 \quad (m \geq 1). \quad (6)$$

This flatness is the source of both its extraordinary endpoint decay and its failure to be analytic there.

2.2 Rvachev's compactly supported bump

Definition 2.2 (Rvachev up-function). Given F , define

$$\text{up}(x) = \begin{cases} F(x+1), & x \leq 0, \\ F(1-x), & x > 0. \end{cases} \quad (7)$$

Equivalently,

$$\text{up}(x) = F(1 - |x|). \quad (8)$$

Here the right side is interpreted using the globally bounded function F .

It follows immediately that up is even, nonnegative, supported on $[-1, 1]$, and $\text{up}(0) = 1$. On the unit interval the two functions are simply translates:

$$F(x) = \text{up}(x-1), \quad 0 \leq x \leq 1. \quad (9)$$

The graph of F is an infinitely flat sigmoid; the graph of up is its symmetric fold.

2.3 The signed global extension

Let $\text{wt}_2(n)$ denote the sum of the binary digits of n , and put

$$\varepsilon_n = (-1)^{\text{wt}_2(n)}. \quad (10)$$

The sequence (ε_n) is the $\{\pm 1\}$ form of the Thue–Morse sequence.

Definition 2.3 (Signed extension). Define

$$\mathcal{F}(x) = \sum_{n \geq 0} \varepsilon_n \text{up}(x - 2n - 1). \quad (11)$$

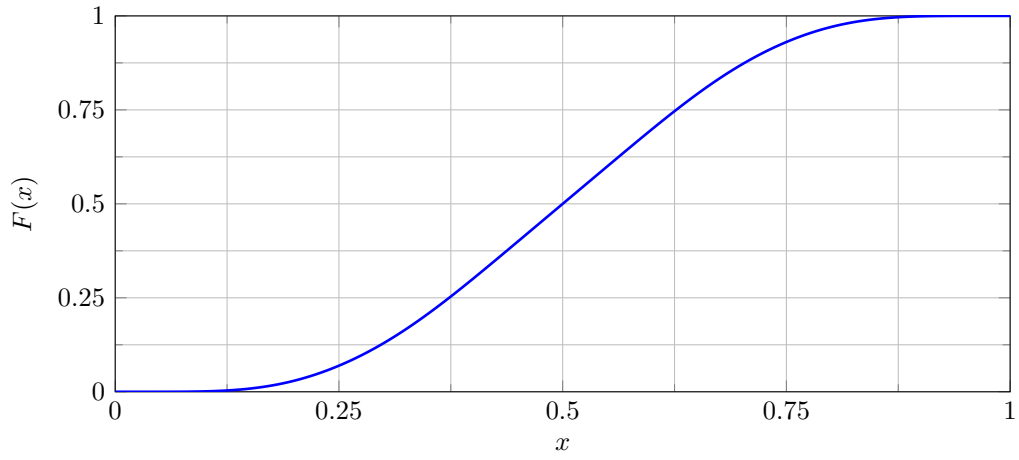


Figure 1: The bounded Fabius function on $[0, 1]$, drawn through exact dyadic values on a 2^{-8} grid. At this resolution the straight segments are visually indistinguishable from the smooth curve.

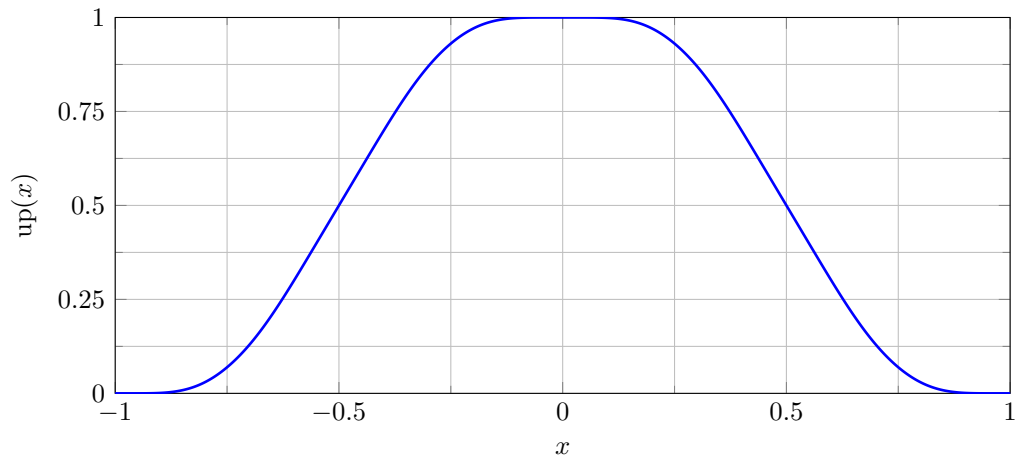


Figure 2: Rvachev's up-function on its support $[-1, 1]$, obtained by folding the same exact dyadic data.

At a fixed x only finitely many summands can be nonzero, so this is a locally finite sum. On every block $[2b, 2b + 2]$ there is exactly one nonzero translate:

$$\mathcal{F}(x) = \varepsilon_b \text{up}(x - 2b - 1), \quad 2b \leq x \leq 2b + 2. \quad (12)$$

It vanishes for $x \leq 0$, and it agrees with the bounded F on $[0, 1]$. It does *not* remain in $[0, 1]$ to the right of that interval. This is the global object that satisfies the cleanest differential equation:

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \quad (x \in \mathbb{R}). \quad (13)$$

Arias de Reyna's arithmetic paper uses this extension. Statements about its values outside $[0, 1]$ must therefore not be transferred blindly to the bounded cumulative distribution.

Notation used in this article. F always means the bounded distribution function; up always means the compactly supported even bump; and $\mathcal{F}$ always means the signed global extension. On $[0, 1]$ we have $F = \mathcal{F}$, but globally they are different functions.

3 Existence and uniqueness by a contraction

The most economical construction of F is a Banach fixed-point argument. It has an additional advantage: the contraction estimate later gives a clean uniqueness theorem, without presupposing Fourier analysis or probability.

3.1 The admissible function space

Let $I = [0, 1]$ and let $C(I)$ carry the supremum norm. Define $\mathcal{A} \subset C(I)$ by

$$\mathcal{A} = \{f \in C(I) : 0 \leq f \leq 1, f(1 - x) = 1 - f(x)\}. \quad (14)$$

The set $\mathcal{A}$ is closed in the Banach space $C(I)$, hence complete. It is nonempty; for example $f(x) = x$ belongs to it.

Extend $f \in \mathcal{A}$ to a continuous function $\tilde{f}$ on $\mathbb{R}$ by clamping the argument to I :

$$\tilde{f}(t) = \begin{cases} f(0), & t \leq 0, \\ f(t), & 0 \leq t \leq 1, \\ f(1), & t \geq 1. \end{cases} \quad (15)$$

Symmetry gives

$$\int_0^1 f(t) dt = \frac{1}{2}. \quad (16)$$

Indeed, replacing t by $1 - t$ and adding the two integrals gives $2 \int_0^1 f = 1$.

3.2 The integral transform

For $f \in \mathcal{A}$, define Tf by

$$(Tf)(x) = \begin{cases} \int_0^{2x} \tilde{f}(t) dt, & 0 \leq x \leq \frac{1}{2}, \\ 1 - \int_0^{2-2x} \tilde{f}(t) dt, & \frac{1}{2} < x \leq 1. \end{cases} \quad (17)$$

The two definitions agree at $x = 1/2$ by (16). Positivity and the upper bound $f \leq 1$ show that $0 \leq Tf \leq 1$, and a direct substitution shows $(Tf)(1 - x) = 1 - (Tf)(x)$. Hence $T : \mathcal{A} \rightarrow \mathcal{A}$.

The central estimate is stronger than the naive integral bound.

Lemma 3.1 (Contraction estimate). *For $f, g \in \mathcal{A}$,*

$$\|Tf - Tg\|_\infty \leq \frac{1}{2}\|f - g\|_\infty. \quad (18)$$

Proof. Write $h = f - g$. Since both f and g have integral $1/2$, $\int_0^1 h = 0$. For $0 \leq y \leq 1$ we may therefore estimate the primitive in two ways:

$$\begin{aligned} \left| \int_0^y h(t) dt \right| &\leq y\|h\|_\infty, \\ \left| \int_0^y h(t) dt \right| &= \left| \int_y^1 h(t) dt \right| \leq (1-y)\|h\|_\infty. \end{aligned}$$

Consequently

$$\left| \int_0^y h(t) dt \right| \leq \min(y, 1-y)\|h\|_\infty \leq \frac{1}{2}\|h\|_\infty.$$

Every value of $Tf - Tg$ is one of these primitives, possibly with a minus sign. Taking the supremum proves (18). $\square$

Theorem 3.2 (Existence of the bounded Fabius function). *The transform T has a unique fixed point $f_* \in \mathcal{A}$. Extending f_* by 0 on $(-\infty, 0]$ and by 1 on $[1, \infty)$ produces the bounded Fabius function F .*

Proof. The space $\mathcal{A}$ is complete and T is a strict contraction with constant $1/2$, so Banach's fixed-point theorem gives a unique fixed point. On $[0, 1/2]$ its fixed-point equation is

$$F(x) = \int_0^{2x} F(t) dt. \quad (19)$$

Differentiating gives $F'(x) = 2F(2x)$. The second half of (17) supplies symmetry and the matching right-hand formula. The endpoint values are immediate. $\square$

Starting from any admissible f_0 , the iterates $f_{n+1} = Tf_n$ converge uniformly to F , with the explicit estimate

$$\|f_n - F\|_\infty \leq 2^{-n}\|f_0 - F\|_\infty. \quad (20)$$

This gives a conceptually simple approximation scheme, although exact dyadic evaluation will be much faster.

3.3 The differential equation for up and smoothness

Folding (4) and (5) yields the refinement-type differential equation

$$\text{up}'(x) = 2(\text{up}(2x+1) - \text{up}(2x-1)) \quad (x \in \mathbb{R}). \quad (21)$$

At $x = 0$ the two one-sided derivatives glue because $\text{up}(1) = \text{up}(-1) = 0$. Once up is continuously differentiable, (21) bootstraps regularity: if $\text{up} \in C^r$, then the right side is C^r , so $\text{up} \in C^{r+1}$. Induction gives $\text{up} \in C^\infty$ and then $F \in C^\infty$.

Proposition 3.3 (Basic shape). *The functions F and up have the following properties.*

1. F is strictly increasing on $(0, 1)$ and satisfies $0 < F(x) < 1$ there.
2. up is even, strictly positive on $(-1, 1)$, and zero outside $[-1, 1]$.
3. $\int_{-1}^1 \text{up}(x) dx = 1$.
4. Every derivative of up vanishes at $x = \pm 1$.

Proof. For $0 < x \leq 1/2$, (4) gives $F'(x) = 2F(2x)$. Positivity follows first near $1/2$ from $F(1/2) = 1/2$, and then propagates toward zero by repeatedly doubling the argument; symmetry handles the right half. This proves strict monotonicity and, by (8), positivity of up in the interior of its support. Finally,

$$\int_{-1}^1 \text{up}(x) dx = 2 \int_0^1 F(t) dt = 1,$$

where the middle identity uses the fold and the last uses symmetry. Flatness at the support endpoints follows from smoothness and the fact that up is identically zero just outside the support. $\square$

3.4 The original Rvachev characterization

The preceding construction is equivalent to the characterization historically associated with Rvachev.

Theorem 3.4 (Original characterization). *There is a unique nonzero C^∞ function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ such that*

$$\text{supp } \varphi = [-1, 1], \quad \int_{\mathbb{R}} \varphi(x) dx = 1, \quad (22)$$

and

$$\varphi'(x) = k(\varphi(2x+1) - \varphi(2x-1)) \quad (23)$$

for some real scale k . Necessarily $k = 2$, and $\varphi = \text{up}$.

Proof architecture. Integrating (23) against suitable test functions, or taking its Fourier transform, first forces $k = 2$ under the normalization (22). With the Fourier convention

$$\widehat{\varphi}(z) = \int_{\mathbb{R}} \varphi(x) e^{-2\pi i x z} dx, \quad (24)$$

one obtains

$$\widehat{\varphi}(z) = \text{sinc}(\pi z) \widehat{\varphi}(z/2), \quad \text{sinc } z = \begin{cases} \sin z/z, & z \neq 0, \\ 1, & z = 0. \end{cases} \quad (25)$$

Iterating N times gives a finite sinc product multiplied by $\widehat{\varphi}(z/2^N)$. Continuity and $\widehat{\varphi}(0) = 1$ determine the Fourier transform uniquely in the limit. Fourier inversion then determines φ . The function constructed above satisfies all the hypotheses, so it is that unique solution. $\square$

4 The probability law behind the functions

The fixed-point construction is analytic. A probabilistic construction explains the same self-similarity almost without calculation and makes monotonicity and positivity transparent.

4.1 An infinite weighted sum of uniforms

Let $U_0, U_1, \dots$ be independent random variables, each uniformly distributed on $[0, 1]$, and define

$$X = \sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}}. \quad (26)$$

The series converges absolutely for every outcome and satisfies $0 \leq X \leq 1$. Removing the first coordinate gives an independent copy X' of X , and

$$X = \frac{U_0 + X'}{2}. \quad (27)$$

Reflecting every coordinate, $U_n \mapsto 1 - U_n$, preserves the product measure and sends X to $1 - X$.

Let $H(x) = \mathbb{P}(X \leq x)$ be the cumulative distribution function. Conditioning on U_0 in (27) gives

$$H(y) = \int_0^1 H(2y - u) \, du. \quad (28)$$

For $0 \leq y \leq 1/2$, the integrand vanishes when $u > 2y$, so

$$H(y) = \int_0^{2y} H(t) \, dt. \quad (29)$$

Reflection gives $H(1 - y) = 1 - H(y)$. Thus the restriction of H to $[0, 1]$ is a fixed point of the contraction T from section 3.

Theorem 4.1 (Probabilistic representation). *For every $x \in \mathbb{R}$,*

$$F(x) = \mathbb{P}\left(\sum_{n=0}^{\infty} \frac{U_n}{2^{n+1}} \leq x\right). \quad (30)$$

For $-1 \leq y \leq 0$,

$$\text{up}(y) = \mathbb{P}(X \leq y + 1). \quad (31)$$

Proof. The preceding calculation shows that H is an admissible fixed point of T . Uniqueness of the fixed point gives $H = F$. Equation (31) is then just the left branch of (7). $\square$

The more symmetric random variable is

$$Y = 2X - 1 = \sum_{n=0}^{\infty} \frac{V_n}{2^{n+1}}, \quad V_n = 2U_n - 1 \sim \text{Unif}[-1, 1]. \quad (32)$$

The density of Y is precisely up . Indeed, for $-1 < y < 1$,

$$f_Y(y) = \frac{1}{2} F'\left(\frac{y+1}{2}\right) = \text{up}(y). \quad (33)$$

This identity is a compact way to remember why up integrates to one.

4.2 Finite convolutions and step approximants

For each n , the summand $V_n/2^{n+1}$ is uniform on $[-2^{-n-1}, 2^{-n-1}]$ and has density

$$b_n(x) = 2^n \mathbf{1}_{[-2^{-n-1}, 2^{-n-1}]}(x). \quad (34)$$

Hence the partial sum

$$Y_N = \sum_{n=0}^N \frac{V_n}{2^{n+1}} \quad (35)$$

has density

$$u_N = b_0 * b_1 * \cdots * b_N. \quad (36)$$

These compactly supported piecewise-polynomial densities converge to up . Weak convergence is immediate from almost-sure convergence of Y_N ; the refinement equation and uniform bounds upgrade the convergence to the strong forms used in the formal development. This constructs up as an infinite convolution of shrinking boxes, a perspective useful in approximation theory and wavelet analysis.

4.3 Centered finite splines and an explicit error bound

The preceding finite random sums also give a completely explicit polynomial approximation to F . Put

$$T_p = \sum_{i=0}^{p-1} \frac{U_i}{2^{i+1}}, \quad C_p(x) = \mathbb{P}\left(T_p \leq x - 2^{-p-1}\right), \quad (37)$$

where $p \geq 1$. The shift by half of the omitted tail is important: it centers the one-sided approximation and improves its error by a factor of two.

Let

$$N_p(x) = \max\left(0, \left\lfloor 2^p x + \frac{1}{2} \right\rfloor\right), \quad \varepsilon_r = (-1)^{\text{wt}_2(r)}, \quad (38)$$

and interpret a sum with $N_p(x) = 0$ as empty. Define

$$S_p(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} \varepsilon_r \left(r - 2^p x + \frac{1}{2}\right)_+^p. \quad (39)$$

Equivalently,

$$S_p(x) = \frac{1}{2^{\binom{p}{2}} p!} \sum_{r \geq 0} \varepsilon_r \left(2^p x - \frac{1}{2} - r\right)_+^p, \quad (40)$$

where only the terms $r < N_p(x)$ can be nonzero and $t_+ = \max(t, 0)$. The cutoff in (38) is a safe natural-number form of the inclusive upper limit

$$0 \leq r \leq \left\lfloor 2^p x - \frac{1}{2} \right\rfloor.$$

It handles the empty case without assigning a negative upper endpoint to a finite sum.

Theorem 4.2 (Centered spline as a finite distribution function). *For $p \geq 1$ and $0 \leq x \leq 1$,*

$$S_p(x) = C_p(x). \quad (41)$$

Consequently $0 \leq S_p(x) \leq 1$ on the unit interval.

Proof. For $p = 1$, $T_1 = U_0/2$, and direct evaluation gives

$$C_1(x) = \max\left(0, \min\left(1, 2x - \frac{1}{2}\right)\right).$$

Formula (40) gives exactly the same broken-linear function: the term $r = 0$ starts at $x = 1/4$, and the term $r = 1$ subtracts the excess after $x = 3/4$.

For the induction step, split off the first uniform coordinate. The finite sums satisfy

$$T_{p+1} \stackrel{d}{=} \frac{U_0 + T'_p}{2},$$

where T'_p is an independent copy of T_p . Conditioning on U_0 and accounting for the midpoint shifts gives the exact smoothing recurrence

$$C_{p+1}(x) = \int_0^1 C_p(2x - u) du. \quad (42)$$

On the spline side, integrate each truncated power in (40). The two binary children $2r$ and $2r + 1$ have opposite Thue–Morse signs, because $\varepsilon_{2r} = \varepsilon_r$ and $\varepsilon_{2r+1} = -\varepsilon_r$. After the change of scale this yields

$$S_{p+1}(x) = \int_0^1 S_p(2x - u) du. \quad (43)$$

The normalization is exactly the one required by

$$\sum_{r=0}^{2^p-1} \varepsilon_r r^p = (-1)^p 2^{\binom{p}{2}} p!,$$

while every lower power sum vanishes. Thus the two recurrences and their common base case prove (41). $\square$

Theorem 4.3 (Quantitative centered approximation). *For $p \geq 1$ and $0 \leq x \leq 1$,*

$$F(x - 2^{-p-1}) \leq S_p(x) \leq F(x + 2^{-p-1}), \quad (44)$$

and hence

$$|S_p(x) - F(x)| \leq 2^{-p}. \quad (45)$$

Proof. Write $X = T_p + R_p$, where the omitted tail satisfies

$$0 \leq R_p = \sum_{i=p}^{\infty} \frac{U_i}{2^{i+1}} \leq 2^{-p}.$$

If $X \leq x - 2^{-p-1}$, then certainly $T_p \leq x - 2^{-p-1}$. Conversely, if $T_p \leq x - 2^{-p-1}$, then $X \leq x + 2^{-p-1}$. Taking probabilities and using (30) and (41) proves (44).

The bounded Fabius function is globally 2-Lipschitz. Indeed, on the open unit interval its derivative is a rescaled value of up (or of the signed extension), whose absolute value is at most 1, and outside the unit interval F is constant. Thus

$$|F(x \pm 2^{-p-1}) - F(x)| \leq 2 \cdot 2^{-p-1} = 2^{-p}.$$

Combining these two inequalities with the sandwich gives (45). $\square$

The bound is uniform on the closed interval and includes both endpoints. It is stronger than mere weak convergence of the finite distributions, and it is the quantitative input for the computability proof in [section 19.6](#).

The same finite formula also has the correct signed global behavior. Binary block translation and midpoint reflection extend the local identification to every $x \geq 0$:

$$|S_p(x)| \leq 1, \quad S_p(x) \longrightarrow \mathcal{F}(x). \quad (46)$$

This global form is the analytic input for the complex-shift and discrete-limit results below; it should not be confused with uniform convergence on the whole half-line.

5 Fourier transform, entire products, and Poisson summation

5.1 The sinc product

We use the Fourier convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(x) e^{-2\pi i x z} dx. \quad (47)$$

The characteristic function of a uniform variable on $[-a, a]$ is $\text{sinc}(2\pi a z)$. Independence in (32) therefore yields the product formula.

Theorem 5.1 (Fourier product). *For every $z \in \mathbb{C}$,*

$$\widehat{\text{up}}(z) = \prod_{n=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^n}\right). \quad (48)$$

The product converges locally uniformly and hence defines an entire function.

Proof. For real z , the characteristic function of the partial sum Y_N is the finite product through $n = N$. Since $Y_N \rightarrow Y$ almost surely and the exponential has modulus one, dominated convergence gives the real identity. On compact subsets of $\mathbb{C}$,

$$\text{sinc } w = 1 - \frac{w^2}{6} + \mathcal{O}(w^4),$$

so the sum of the deviations of the factors from 1 is dominated by $C_K \sum 4^{-n}$. The Weierstrass product theorem gives local uniform convergence to an entire function. Equality on the real axis and the identity theorem extend the formula to all complex z . $\square$

Separating the first factor gives the refinement equation already seen in (25):

$$\widehat{\text{up}}(z) = \text{sinc}(\pi z) \widehat{\text{up}}(z/2). \quad (49)$$

Conversely, continuity at zero, normalization $\widehat{\text{up}}(0) = 1$, and (49) force (48) by iteration. This is the core of the Fourier uniqueness proof.

5.2 Rapid decay and inversion

Since $\text{up} \in C_c^\infty(\mathbb{R})$, repeated integration by parts gives, for every integer $N \geq 0$,

$$|\widehat{\text{up}}(\xi)| \leq C_N (1 + |\xi|)^{-N}. \quad (50)$$

Thus the Fourier product is rapidly decreasing on the real axis despite being entire of finite exponential type. Fourier inversion is absolutely convergent:

$$\text{up}(x) = \int_{\mathbb{R}} \widehat{\text{up}}(\xi) e^{2\pi i x \xi} d\xi. \quad (51)$$

5.3 Poisson summation and partitions of unity

Rapid decay on both sides makes Poisson summation available without distributional qualifications.

Theorem 5.2 (Poisson summation). *For $u > 0$ and $t \in \mathbb{R}$,*

$$\sum_{k \in \mathbb{Z}} \text{up}(t + uk) = \frac{1}{u} \sum_{k \in \mathbb{Z}} \widehat{\text{up}}(k/u) e^{2\pi i kt/u}. \quad (52)$$

Both series converge absolutely and locally uniformly.

For a positive integer m , every nonzero term $\widehat{\text{up}}(mk)$ vanishes because the first sinc factor in (48) vanishes at nonzero integers. Taking $u = 1/m$ in (52) gives the exact partition of unity

$$\sum_{k \in \mathbb{Z}} \text{up}\left(t + \frac{k}{m}\right) = m. \quad (53)$$

In particular,

$$\sum_{k \in \mathbb{Z}} \text{up}(t + k) = 1. \quad (54)$$

This identity is one reason up is useful as a smooth compactly supported averaging kernel.

5.4 Moment expansion of the Fourier transform

Define the even moments

$$c_n = \int_{-1}^1 x^{2n} \text{up}(x) \, dx. \quad (55)$$

Evenness eliminates the odd moments, and expansion of the exponential in (47) gives the entire series

$$\widehat{\text{up}}(z) = \sum_{n=0}^{\infty} \frac{(-1)^n c_n}{(2n)!} (2\pi z)^{2n}. \quad (56)$$

Comparing this with the sinc refinement will produce the exact rational recurrence for c_n in [section 7](#).

6 The global extension and exact derivative identities

The signed extension $\mathcal{F}$ is not merely a device for shortening notation. It turns the piecewise differential structure of F and up into a global scaling law, and from that law one can read off all higher derivatives.

6.1 Thue–Morse splitting

The binary weight satisfies

$$\text{wt}_2(2n) = \text{wt}_2(n), \quad \text{wt}_2(2n+1) = \text{wt}_2(n) + 1. \quad (57)$$

Therefore

$$\varepsilon_{2n} = \varepsilon_n, \quad \varepsilon_{2n+1} = -\varepsilon_n. \quad (58)$$

Differentiate the locally finite sum (11) term by term and use (21). The result is a sum over even and odd indices. Reindexing those two subseries with (58) makes them recombine into $2\mathcal{F}(2x)$.

Theorem 6.1 (Global differential equation). *The signed extension is infinitely differentiable and*

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \quad (59)$$

for every real x .

Proof. Local finiteness permits differentiation of (11). By (21),

$$\begin{aligned} \mathcal{F}'(x) &= 2 \sum_{n \geq 0} \varepsilon_n [\text{up}(2x - 4n - 1) - \text{up}(2x - 4n - 3)] \\ &= 2 \sum_{n \geq 0} \varepsilon_{2n} \text{up}(2x - 2(2n) - 1) + 2 \sum_{n \geq 0} \varepsilon_{2n+1} \text{up}(2x - 2(2n+1) - 1) \\ &= 2 \sum_{m \geq 0} \varepsilon_m \text{up}(2x - 2m - 1) = 2\mathcal{F}(2x). \end{aligned}$$

The same local-finiteness argument shows that the sum is C^∞ because each translate of up is. □

6.2 All iterated derivatives

Repeated differentiation of (59) gives a triangular power of two.

Theorem 6.2 (Derivative scaling). *For every integer $m \geq 0$,*

$$\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x). \quad (60)$$

For $-1 \leq x \leq 1$,

$$\text{up}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x + 2^m). \quad (61)$$

Proof. The first assertion is immediate by induction. If it holds for m , then

$$\begin{aligned} \mathcal{F}^{(m+1)}(x) &= 2^{\binom{m+1}{2}} 2^m \mathcal{F}'(2^m x) \\ &= 2^{\binom{m+1}{2}} 2^m \cdot 2 \mathcal{F}(2^{m+1} x) \\ &= 2^{\binom{m+2}{2}} \mathcal{F}(2^{m+1} x). \end{aligned}$$

On $[-1, 1]$, $\text{up}(x) = \mathcal{F}(x + 1)$, so $\text{up}^{(m)}(x) = \mathcal{F}^{(m)}(x + 1)$ and the second formula follows. $\square$

The formula is striking: the m th derivative is not an approximation or an asymptotic rescaling of the original function; it is an exact rescaling of the signed extension. The large factor $2^{m(m+1)/2}$ quantifies the rapid growth of high derivatives, while the argument $2^m x$ accounts for the increasingly fine oscillatory structure.

6.3 Polynomial behavior near dyadic points

Let $x_0 = a/2^n$ be a dyadic point in the interior of the unit interval. For $m > n$, the argument $2^m x_0$ in (60) is an even integer. The local translate formula (12) and flatness of up at its endpoints imply

$$F^{(m)}(x_0) = 0 \quad (m > n). \quad (62)$$

Thus the Taylor series of F at a dyadic point terminates.

Proposition 6.3 (Terminating Taylor series). *If $x_0 = a/2^n \in (0, 1)$ with a odd, then the formal Taylor series of F at x_0 is a polynomial of degree at most n . Nevertheless F does not agree with that polynomial on any neighborhood of x_0 .*

Proof. Equation (62) proves termination. If F agreed with the Taylor polynomial on a nonempty interval, analyticity of the polynomial and the differential scaling relation would propagate polynomial behavior across successively enlarged dyadic intervals. Ultimately F would be polynomial on $(0, 1)$, which is incompatible with its flatness at 0 and its nonconstant value $F(1/2) = 1/2$. Equivalently, a polynomial with all positive-order derivatives zero at 0 is constant, whereas F is not. $\square$

Corollary 6.4 (Nowhere analytic). *Neither F on $(0, 1)$ nor up on $(-1, 1)$ is real analytic at any point.*

Proof. Dyadic points are dense. If F were analytic at one point, the differential equations and translation/reflection identities would extend analyticity to an interval and then to dyadic points in that interval, contradicting the preceding proposition. The statement for up follows from the fold relation. $\square$

7 Moments and generating functions

Exact arithmetic enters through two rational moment sequences. One is attached to the even moments of up; the other records its one-sided moments and, at the same time, the values of F at reciprocal powers of two.

7.1 The even moments c_n

Recall

$$c_n = \int_{-1}^1 x^{2n} \text{up}(x) \, dx, \quad c_0 = 1.$$

Theorem 7.1 (Moment recurrence). *For $n \geq 1$,*

$$(2n+1)(2^{2n}-1)c_n = \sum_{k=0}^{n-1} \binom{2n+1}{2k} c_k. \quad (63)$$

Consequently every c_n is rational and positive.

Proof. Insert the power series (56) into (49). Since

$$\text{sinc}(\pi z) = \sum_{j \geq 0} \frac{(-1)^j (\pi z)^{2j}}{(2j+1)!}$$

and

$$\widehat{\text{up}}(z/2) = \sum_{k \geq 0} \frac{(-1)^k c_k (\pi z)^{2k}}{(2k)!},$$

comparison of the coefficient of $(-1)^n (\pi z)^{2n}$ gives

$$\frac{2^{2n} c_n}{(2n)!} = \sum_{k=0}^n \frac{c_k}{(2k)!(2n-2k+1)!}.$$

Multiplying by $(2n+1)!$ yields

$$(2n+1)2^{2n} c_n = \sum_{k=0}^n \binom{2n+1}{2k} c_k.$$

The term $k=n$ is $(2n+1)c_n$; moving it to the left gives (63). Positivity follows either from the integral definition or by induction from the recurrence. $\square$

The first values are

$$1, \quad \frac{1}{9}, \quad \frac{19}{675}, \quad \frac{583}{59535}, \quad \frac{132809}{32531625}, \dots \quad (64)$$

A useful denominator clearing is

$$c_n = \frac{C_n}{(2n+1)!! \prod_{j=1}^n (2^{2j}-1)}, \quad (65)$$

where C_n is a positive integer. It can be defined without division by

$$C_0 = 1, \quad (66)$$

$$C_{n+1} = \sum_{k=0}^n C_k \binom{2n+3}{2k} \prod_{j=k+1}^n (2j+1) \prod_{j=k+1}^n (2^{2j}-1). \quad (67)$$

The formal development proves directly that every C_n is odd. Modulo 2, all factors in the two products are 1, and Lucas's theorem shows that among the coefficients $\binom{2n+3}{2k}$ with $0 \leq k \leq n$ an odd number are odd. This closes the induction.

7.2 The half moments d_n

Define

$$d_0 = 1, \quad d_n = n \int_0^1 t^{n-1} \text{up}(t) dt \quad (n \geq 1). \quad (68)$$

Introduce the entire exponential generating function

$$G(z) = 1 + z \int_0^1 \text{up}(t) e^{zt} dt = \sum_{n=0}^{\infty} d_n \frac{z^n}{n!}. \quad (69)$$

Theorem 7.2 (Generating-function equation). *For every $z \in \mathbb{C}$,*

$$G(2z) = \frac{e^z - 1}{z} G(z), \quad (70)$$

where the quotient has its removable value 1 at $z = 0$.

Proof. One may derive the identity directly from (21) by integration by parts. An especially transparent route uses Fourier analysis. From evenness and the fold relation,

$$G(z) = e^{z/2} \widehat{\text{up}}\left(\frac{iz}{4\pi}\right). \quad (71)$$

Applying (49) at $iz/(2\pi)$ and simplifying $\text{sinc}(iz/2) = 2 \sinh(z/2)/z$ gives

$$G(2z) = e^z \frac{2 \sinh(z/2)}{z} \widehat{\text{up}}\left(\frac{iz}{4\pi}\right) = \frac{e^z - 1}{z} G(z).$$

□

Corollary 7.3 (Half-moment recurrence). *For $n \geq 1$,*

$$(n+1)(2^n - 1)d_n = \sum_{k=0}^{n-1} \binom{n+1}{k} d_k. \quad (72)$$

In particular $d_n \in \mathbb{Q}$.

Proof. Expand $(e^z - 1)/z = \sum_{j \geq 0} z^j/(j+1)!$ in (70). Comparison of the coefficient of $z^n/n!$ gives

$$2^n d_n = \frac{1}{n+1} \sum_{k=0}^n \binom{n+1}{k} d_k.$$

Isolating the $k = n$ term proves the recurrence. □

The first values are

$$1, \quad \frac{1}{2}, \quad \frac{5}{18}, \quad \frac{1}{6}, \quad \frac{143}{1350}, \quad \frac{19}{270}, \quad \frac{1153}{23814}, \dots \quad (73)$$

As for c_n , one may clear denominators without division:

$$d_n = \frac{D_n}{(n+1)! \prod_{j=1}^n (2^j - 1)}, \quad (74)$$

where $D_n \in \mathbb{N}$. A division-free recurrence follows by substituting (74) into (72). The normalized parity theorem says that, for $n \geq 1$, the reduced numerator and denominator of $2d_n$ are both odd.

7.3 Relation between the two moment sequences

Expanding (71) in powers of z gives a finite binomial transform at each order.

Proposition 7.4. *For every $n \geq 0$,*

$$d_n = 2^{-n} \sum_{0 \leq k \leq n/2} \binom{n}{2k} c_k. \quad (75)$$

Proof. By (56),

$$\widehat{\text{up}}\left(\frac{iz}{4\pi}\right) = \sum_{k \geq 0} c_k \frac{(z/2)^{2k}}{(2k)!}.$$

Multiplying by $e^{z/2}$ and extracting the coefficient of $z^n/n!$ gives exactly (75). $\square$

7.4 Bernoulli-number recurrences

Taking logarithms in (70) and using

$$\log \frac{e^z - 1}{z} = \frac{z}{2} + \sum_{m \geq 1} \frac{B_{2m}}{2m(2m)!} z^{2m} \quad (76)$$

produces recurrences in which Bernoulli numbers replace binomial convolutions. One convenient form for the even moments is obtained by taking the logarithmic derivative of the sinc product. If

$$\widehat{\text{up}}(z) = \sum_{n \geq 0} (-1)^n c_n \frac{(2\pi z)^{2n}}{(2n)!},$$

then the coefficients satisfy a triangular recurrence whose m th logarithmic coefficient is proportional to

$$\frac{B_{2m}}{m(2m)!} \frac{1}{1 - 2^{-2m}}. \quad (77)$$

The formal development proves both the ordinary convolution recurrences and these Bernoulli forms, and proves their equivalence by formal power-series algebra. The binomial recurrences are usually better for exact computation; the Bernoulli forms are often better for asymptotic estimates.

8 Exact values at reciprocal powers of two

The half moments are already endpoint values in disguise.

Theorem 8.1 (Reciprocal-power values). *For every $n \geq 0$,*

$$F(2^{-n}) = \frac{d_n}{n! 2^{\binom{n}{2}}}. \quad (78)$$

Equivalently,

$$d_n = n! 2^{\binom{n}{2}} F(2^{-n}). \quad (79)$$

Proof. For $n \geq 1$, repeated integration by parts using a chain of primitives generated by (21) gives

$$\int_0^1 t^{n-1} \text{up}(t) dt = (n-1)! 2^{\binom{n}{2}} F(2^{-n}). \quad (80)$$

Multiplying by n and using (68) proves the formula. The case $n = 0$ is $F(1) = d_0 = 1$. $\square$

The first values are therefore

n	$F(2^{-n})$	
0	1	
1	$\frac{1}{2}$	
2	$\frac{5}{72}$	
3	$\frac{288}{143}$	
4	$\frac{2073600}{19}$	
5	$\frac{33177600}{1153}$	
6	$\frac{561842749440}{583}$	
7	$\frac{179789679820800}{1}$	(81)

The denominators grow far faster than exponentially, in agreement with the quadratic leading term in the logarithmic asymptotic derived later.

8.1 A triangular recurrence and its nonrecursive solution

The values in (81) may be generated without first computing the moments. Write

$$V_n = F(2^{-n}). \quad (82)$$

Eliminating d_n between (72) and (78) gives, for $n \geq 1$,

$$V_n = \frac{2^{-\binom{n}{2}}}{2^n - 1} \sum_{k=0}^{n-1} \frac{2^{\binom{k}{2}}}{(n-k+1)!} V_k, \quad V_0 = 1. \quad (83)$$

The restriction $n \geq 1$ is essential: at $n = 0$ the denominator $2^n - 1$ is zero and the sum is empty, whereas $V_0 = F(1) = 1$. Thus the initial value is part of the recurrence data, not a removable endpoint convention.

The triangular powers of two disappear after the normalization

$$A_n = 2^{\binom{n}{2}} V_n. \quad (84)$$

Indeed, $A_0 = 1$ and

$$A_n = \sum_{k=0}^{n-1} \frac{A_k}{(2^n - 1)(n - k + 1)!}. \quad (85)$$

This is a weighted-path recurrence on the directed graph with one edge $k \rightarrow n$ whenever $k < n$. Iterating it terminates, and the resulting paths can be indexed by ordered compositions.

The recurrence is triangular: once $A_0 = 1$ is fixed, each A_n is a linear combination of earlier values. Induction therefore gives uniqueness. In particular, the argument below is purely algebraic and applies to any sequence satisfying (85); no further analytic property of F is used.

An ordered composition of n is a tuple $\rho = (r_1, \dots, r_m)$ of positive integers with sum n . Put

$$s_j = r_1 + \dots + r_j, \quad W(\rho) = \prod_{j=1}^m \frac{1}{(2^{s_j} - 1)(r_j + 1)!}. \quad (86)$$

For $n = 0$ there is one empty composition; its empty product is 1.

Theorem 8.2 (Finite composition formula). *For every $n \geq 0$,*

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{\rho \models n} \prod_{j=1}^{\text{len}(\rho)} \frac{1}{(2^{s_j} - 1)(r_j + 1)!} \quad (87)$$

where $\rho = (r_1, \dots, r_m)$ and $s_j = r_1 + \dots + r_j$. For $n = 0$ the unique empty composition makes the right side equal to 1.

Proof. Let $C_n = \sum_{\rho \models n} W(\rho)$. The empty-composition convention gives $C_0 = 1$. Fix $n \geq 1$ and split every composition of n immediately before its last block. If the preceding blocks sum to k , then $0 \leq k < n$, the last block is $n - k$, and the last factor of (86) is

$$\frac{1}{(2^n - 1)(n - k + 1)!}.$$

Conversely, appending the block $n - k$ to any composition of k gives exactly one composition of n with penultimate partial sum k . Therefore

$$C_n = \sum_{k=0}^{n-1} \frac{C_k}{(2^n - 1)(n - k + 1)!}.$$

Thus (C_n) and (A_n) have the same initial value and the same triangular recurrence. Induction gives $C_n = A_n$, and multiplying by $2^{-\binom{n}{2}}$ proves (87). $\square$

Every point 2^{-n} lies in $[0, 1]$, where the bounded Fabius function and the signed global extension agree. Consequently the same formula holds with $F(2^{-n})$ replaced by $\mathcal{F}(2^{-n})$.

Equivalently, put $i_0 = 0$ and $i_j = s_j$. Compositions of length m correspond bijectively to chains

$$0 = i_0 < i_1 < \dots < i_m = n, \quad r_j = i_j - i_{j-1}.$$

Hence, for $n \geq 1$,

$$F(2^{-n}) = 2^{-\binom{n}{2}} \sum_{m=1}^n \sum_{0=i_0 < i_1 < \dots < i_m=n} \prod_{j=1}^m \frac{1}{(2^{i_j} - 1)(i_j - i_{j-1} + 1)!}. \quad (88)$$

There are 2^{n-1} compositions of a positive integer n , so this is finite and nonrecursive, although exponential in size. For fixed m , the inner sum may instead be written over $1 \leq i_1 < \dots < i_{m-1} \leq n - 1$, with $i_0 = 0$ and $i_m = n$; when $m = 1$ there are no internal indices and there is one direct path $0 \rightarrow n$.

The common factor $2^{-\binom{n}{2}}$ has a direct path interpretation. Before normalization, the edge $k \rightarrow r$ has weight

$$U_{r,k} = \frac{2^{\binom{k}{2} - \binom{r}{2}}}{(2^r - 1)(r - k + 1)!}. \quad (89)$$

Along a chain $0 = i_0 < i_1 < \dots < i_m = n$, the powers of two telescope:

$$\prod_{j=1}^m 2^{\binom{i_{j-1}}{2} - \binom{i_j}{2}} = 2^{\binom{i_0}{2} - \binom{i_m}{2}} = 2^{-\binom{n}{2}}. \quad (90)$$

The remaining factors are precisely the normalized chain weight in (88). Thus normalization extracts an endpoint factor common to every path, rather than merely shortening the recurrence.

Example 8.3 (The first nontrivial path sums). For $n = 2$, the compositions are (2) and $(1, 1)$, with weights

$$W(2) = \frac{1}{(2^2 - 1)3!} = \frac{1}{18}, \quad W(1, 1) = \frac{1}{(2^1 - 1)2!} \frac{1}{(2^2 - 1)2!} = \frac{1}{12}.$$

Hence $A_2 = 5/36$ and $F(2^{-2}) = 2^{-1}A_2 = 5/72$. For $n = 3$, the four ordered compositions $(3), (1, 2), (2, 1), (1, 1, 1)$ have weights

$$\frac{1}{168}, \quad \frac{1}{84}, \quad \frac{1}{252}, \quad \frac{1}{168}.$$

The unequal weights of $(1, 2)$ and $(2, 1)$ record their different intermediate partial sums. Their total is $A_3 = 1/36$, giving $F(2^{-3}) = 2^{-3}A_3 = 1/288$.

For a precise matrix form, fix N and define a matrix indexed by $0, \dots, N$ by

$$T_{i,j}^{[N]} = \begin{cases} \frac{1}{(2^i - 1)(i - j + 1)!}, & j < i, \\ 0, & j \geq i. \end{cases}$$

With $\mathbf{A} = (A_0, \dots, A_N)^\top$ and $\mathbf{e}_0 = (1, 0, \dots, 0)^\top$, the initial value and recurrence become

$$\mathbf{A} = \mathbf{e}_0 + T^{[N]} \mathbf{A}.$$

Since $T^{[N]}$ is strictly lower triangular, $(T^{[N]})^{N+1} = 0$ and

$$\mathbf{A} = (I - T^{[N]})^{-1} \mathbf{e}_0 = (I + T^{[N]} + \dots + (T^{[N]})^N) \mathbf{e}_0.$$

The entry $(T^{[N]})_{n,0}^m$ is the sum of products of edge weights over all length- m increasing paths from 0 to n , so its n th component is exactly (88). This is a finite geometric identity for a nilpotent matrix, not an analytic convergence argument. Finally, if

$$D = \text{diag}(2^{\binom{0}{2}}, 2^{\binom{1}{2}}, \dots, 2^{\binom{N}{2}}),$$

then the matrix of the original recurrence is $U = D^{-1}T^{[N]}D$. This diagonal similarity is the matrix form of (90).

8.2 An exact two-adic valuation

The parity theorem for d_n immediately determines the power of two in the reduced denominator.

Theorem 8.4 (Exact 2-adic valuation). *For $n \geq 1$,*

$$v_2(F(2^{-n})) = -\binom{n}{2} - 1 - v_2(n!). \quad (91)$$

Proof. The reduced numerator and denominator of $2d_n$ are odd, so $v_2(d_n) = -1$. Apply (78) and add valuations:

$$v_2(F(2^{-n})) = v_2(d_n) - v_2(n!) - \binom{n}{2}.$$

□

Legendre's formula

$$v_2(n!) = n - \text{wt}_2(n) \quad (92)$$

then makes the binary dependence explicit.

8.3 A normalized odd integer

For $n \geq 1$ define

$$R_n = 2^{\binom{n-1}{2}} (2n)! F(2^{-n}) \prod_{j=1}^{\lfloor n/2 \rfloor} (2^{2j} - 1). \quad (93)$$

The exponent of two in (93) is *positive*; a negative sign that appears in an abstract of the source paper is a typographical error.

Theorem 8.5. *Each R_n is an odd positive integer. Moreover it has a finite expression in the moment numerators C_k and the odd Mersenne products.*

Proof architecture. Insert (75) into (78), then substitute the normalized form (65). After factorial and double-factorial cancellations, every term has an integral common denominator that is killed by the product in (93). Modulo 2, all Mersenne and odd-factor products are 1, and Lucas's theorem reduces the sum to an odd count of odd binomial coefficients. Thus the resulting integer is odd. $\square$

9 Exact evaluation at arbitrary dyadic rationals

The rationality theorem for dyadic arguments is much stronger than the statement “ F happens to take rational values there.” It comes with two complementary exact algorithms:

1. a closed finite formula involving Thue–Morse signs and the rational moments c_k ;
2. a bit-recursive Taylor algorithm whose number of recursive calls is the number of set bits in the numerator.

The first is best for proofs; the second is best for computation.

9.1 The closed Thue–Morse formula

Theorem 9.1 (Exact dyadic formula). *Let $n, a \in \mathbb{N}$ with $0 \leq a \leq 2^n$. Then*

$$F\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{a-1} (-1)^{\text{wt}_2(h)} \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} c_k (2a - 2h - 1)^{n-2k}. \quad (94)$$

An empty outer sum has value 0.

Proof. Write $x = a/2^n$ and use $F(x) = \text{up}(x - 1)$. Since up and all its derivatives vanish at -1 , Taylor's theorem with integral remainder gives

$$\text{up}(x - 1) = \frac{1}{n!} \int_{-1}^{x-1} (x - 1 - t)^n \text{up}^{(n+1)}(t) dt. \quad (95)$$

Insert the exact derivative identity (61):

$$\text{up}^{(n+1)}(t) = 2^{\binom{n+2}{2}} \mathcal{F}(2^{n+1}(t + 1)).$$

Put $s = 2^{n+1}(t + 1)$. As t runs from -1 to $x - 1$, s runs from 0 to $2a$. Split this interval into the blocks $[2h, 2h + 2]$, $0 \leq h < a$. On the h th block, (12) gives

$$\mathcal{F}(s) = (-1)^{\text{wt}_2(h)} \text{up}(s - 2h - 1).$$

The substitution $u = s - 2h - 1$ maps the block to $[-1, 1]$. Gathering the powers of two gives

$$F(a/2^n) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{a-1} (-1)^{\text{wt}_2(h)} \int_{-1}^1 (2a - 2h - 1 - u)^n \text{up}(u) du. \quad (96)$$

Expand the n th power. Odd powers of u integrate to zero because up is even, while

$$\int_{-1}^1 u^{2k} \text{up}(u) du = c_k.$$

The surviving even terms are exactly the inner sum in (94). $\square$

Every quantity on the right of (94) is rational, so dyadic rationality is immediate. More importantly, the formula lives entirely in $\mathbb{Q}$ and can be evaluated without roundoff.

9.2 Thue–Morse cancellation

The mechanism that makes the dyadic formulas simplify is the Prouhet cancellation carried by the Thue–Morse signs.

Lemma 9.2 (Affine power cancellation). *If $d < b$, then for all $X, Y \in \mathbb{Q}$,*

$$\sum_{h=0}^{2^b-1} (-1)^{\text{wt}_2(h)} (X + Yh)^d = 0. \quad (97)$$

Proof. It is enough to prove the claim for h^j , $0 \leq j \leq d$, and then expand the affine power. Let

$$S_{b,j} = \sum_{h=0}^{2^b-1} (-1)^{\text{wt}_2(h)} h^j.$$

Split the sum for $S_{b+1,j}$ at 2^b and use $\varepsilon_{2^b+h} = -\varepsilon_h$:

$$\begin{aligned} S_{b+1,j} &= \sum_{h < 2^b} \varepsilon_h (h^j - (2^b + h)^j) \\ &= - \sum_{r=0}^{j-1} \binom{j}{r} 2^{b(j-r)} S_{b,r}. \end{aligned}$$

If $j < b + 1$, every index r on the right satisfies $r < b$, so the induction hypothesis makes all terms zero. The base case is immediate. $\square$

A compact generating-function version is

$$\sum_{h < 2^b} (-1)^{\text{wt}_2(h)} e^{ht} = \prod_{j=0}^{b-1} (1 - e^{2^j t}), \quad (98)$$

whose right side has a zero of order b at $t = 0$. Differentiating fewer than b times at zero gives the same cancellation.

This lemma explains why apparently large translated sums in (94) collapse to short Taylor blocks when the numerator crosses a binary boundary.

9.3 A Taylor block that removes the leading bit

Suppose $0 < x < 1$, and let r be the unique integer such that

$$2^{-r} \leq x < 2^{-r+1}. \quad (99)$$

Put $y = x - 2^{-r}$. Then $0 \leq y < 2^{-r}$.

Theorem 9.3 (Exact Taylor-block recursion). *With the notation above,*

$$F(x) = -F(y) + \sum_{j=0}^r \frac{2^{\binom{j+1}{2}} F(2^{j-r})}{j!} y^j. \quad (100)$$

Proof. Apply Taylor's theorem at the point 2^{-r} through degree r :

$$F(2^{-r} + y) = \sum_{j=0}^r \frac{F^{(j)}(2^{-r})}{j!} y^j + R_r(y). \quad (101)$$

The derivative scaling formula gives

$$F^{(j)}(2^{-r}) = 2^{\binom{j+1}{2}} F(2^{j-r}), \quad 0 \leq j \leq r. \quad (102)$$

It remains to identify the integral remainder. The scaling equation carries the interval $[2^{-r}, 2^{-r} + y]$ into the adjacent signed translate of the interval $[0, y]$; on that translate the Thue–Morse sign is negative. Reversing the substitutions in the derivative identity shows

$$R_r(y) = -\frac{1}{r!} \int_0^y (y-t)^r F^{(r+1)}(t) dt = -F(y),$$

where the last equality is Taylor's integral formula at the flat point 0. Substitution in (101) proves (100). $\square$

Using (78), the coefficients may be written entirely in terms of the half moments:

$$\frac{2^{\binom{j+1}{2}} F(2^{j-r})}{j!} = \frac{2^{\binom{j+1}{2} - \binom{r-j}{2}} d_{r-j}}{j!(r-j)!}. \quad (103)$$

This is the form that appears in the original arithmetic recurrence.

9.4 The bit-recursive evaluator

Let $x = a/2^e$ with $0 < a < 2^e$. Write

$$b = \lfloor \log_2 a \rfloor, \quad r = e - b, \quad q = a - 2^b. \quad (104)$$

Then 2^{-r} is the leading binary term of x and

$$y = \frac{q}{2^e}. \quad (105)$$

Applying (100) reduces $F(a/2^e)$ to $F(q/2^e)$; the highest set bit of the numerator has disappeared.

Algorithm 9.4 (Exact dyadic evaluator). Precompute $V_j = F(2^{-j})$ for $0 \leq j \leq e$ from (78). Define $\text{Eval}(e, a)$ recursively by

$$\text{Eval}(e, a) = \begin{cases} 0, & a \leq 0, \\ 1, & a \geq 2^e, \\ T_r(q/2^e) - \text{Eval}(e, q), & 0 < a < 2^e, \end{cases}$$

where b, r, q are given by (104) and

$$T_r(z) = \sum_{j=0}^r \frac{2^{\binom{j+1}{2}} V_{r-j}}{j!} z^j. \quad (106)$$

Evaluate T_r by Horner's rule.

A literal pseudocode version is:

```

Eval(e, a, V):
    if a <= 0:      return 0
    if a >= 2^e:    return 1

    b = floor_log2(a)
    r = e - b
    q = a - 2^b
    y = q / 2^e

    p = V[0]                // highest Taylor coefficient, j = r
    for m = 1 .. r:
        j = r - m
        p = V[m] + (2^(j+1) * y / (j+1)) * p

    return p - Eval(e, q, V)
    
```

The displayed loop is one convenient Horner ordering; any exact evaluation of (106) is equivalent.

Theorem 9.5 (Correctness and termination). *For every $e \in \mathbb{N}$ and $a \in \mathbb{Z}$,*

$$\text{Eval}(e, a) = F(a/2^e) \quad (107)$$

under the bounded convention $F = 0$ to the left and $F = 1$ to the right. The recursion makes exactly $\text{wt}_2(a)$ nontrivial calls when $0 \leq a < 2^e$.

Proof. The boundary clauses agree with (1) and (2). In the interior, (100) is precisely the recursive clause. Since $0 \leq q < a$, strong induction proves correctness and termination. Replacing a by q clears its highest set bit and leaves all lower bits unchanged, so the number of nontrivial steps is the popcount $\text{wt}_2(a)$. $\square$

Precomputing the reciprocal-power table costs $O(e^2)$ rational operations using the triangular recurrence for d_n . A single value then costs $O(e \text{wt}_2(a))$ arithmetic operations in a straightforward implementation. This is much smaller than the a outer summands in (94) when a is large.

Example 9.6 (The value at $5/16$). Here $a = 5 = 2^2 + 1$, $e = 4$, so $r = 2$, $q = 1$, and $y = 1/16$. Formula (100) gives

$$\begin{aligned} F(5/16) &= F(1/4) + 2F(1/2)y + \frac{2^3 F(1)}{2} y^2 - F(1/16) \\ &= \frac{5}{72} + \frac{1}{16} + \frac{1}{64} - \frac{143}{2073600} \\ &= \boxed{\frac{305857}{2073600}}. \end{aligned}$$

Only two recursive calls are needed, corresponding to the two set bits of 5.

9.5 A complete level-4 table

The exact values at denominator 16 are shown below. The second half follows from $F(1 - x) = 1 - F(x)$.

x	$F(x)$	x	$F(x)$
0	0	1/2	1/2
1/16	143/2073600	9/16	1295857/2073600
1/8	1/288	5/8	215/288
3/16	46657/2073600	11/16	1767743/2073600
1/4	5/72	3/4	67/72
5/16	305857/2073600	13/16	2026943/2073600
3/8	73/288	7/8	287/288
7/16	777743/2073600	15/16	2073457/2073600

9.6 Denominator bounds

Let

$$M_m = \prod_{j=1}^m (2^{2j} - 1), \quad (2m + 1)!! = 1 \cdot 3 \cdots (2m + 1). \quad (108)$$

Theorem 9.7 (Uniform denominator multiplier). *Let $n \geq 1$ and $a \in \mathbb{Z}$ with $|a| < 2^n$. Then*

$$\text{up}(a/2^n) \, n! \, 2^{\binom{n+1}{2}} (2 \lfloor n/2 \rfloor + 1)!! \, M_{\lfloor n/2 \rfloor} \quad (109)$$

is an integer.

Proof. Apply the global dyadic form of (94). Replace each c_k by (65). For $k \leq \lfloor n/2 \rfloor$, both the odd double factorial in the denominator of c_k and its even-Mersenne product divide the two common products in (109). The remaining factors are integers. Thus every summand becomes integral after multiplication by the displayed common factor. $\square$

Let Δ_n be the least common multiple of the reduced denominators of the values $\text{up}(a/2^n)$ with odd a in the support. The theorem gives the explicit divisibility bound

$$\Delta_n \mid n! \, 2^{\binom{n+1}{2}} (2 \lfloor n/2 \rfloor + 1)!! \, M_{\lfloor n/2 \rfloor}. \quad (110)$$

The exact closed form of Δ_n is subtler. The source paper records a conjectural factorization; the formal development correctly retains it as a conjecture rather than a theorem.

9.7 A q -binomial form, double cancellation, and scalar translation

The formal development derives a second exact dyadic formula organized by Gaussian binomial coefficients. Put

$$(a; q)_n = \prod_{j=0}^{n-1} (1 - aq^j), \quad (111)$$

and write $\begin{bmatrix} n \\ k \end{bmatrix}_q$ for the Gaussian or q -binomial coefficient. The base $q = 1/2$ is forced by the dilation grid $1, 2, 4, \dots$: the corresponding Pochhammer product is the node polynomial for reciprocal dyadic scales.

Theorem 9.8 (Global q -binomial dyadic formula). *For $m, n \in \mathbb{N}$ and every scalar translation $Q \in \mathbb{C}$,*

$$\begin{aligned} \mathcal{F}(m/2^n) &= \frac{1}{2^{n^2}(1/2; 1/2)_n} \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{4^{\binom{k}{2}}(n+k)!} \\ &\times \sum_{r=0}^{m2^k-1} (-1)^{\text{wt}_2(r)} (r - m2^k + Q)^{n+k}. \end{aligned} \quad (112)$$

Here the real value on the left is embedded in $\mathbb{C}$. The right side is independent of Q , so the identity also holds for real, rational, and Gaussian-rational shifts. If $0 \leq m \leq 2^n$, then $\mathcal{F}(m/2^n) = F(m/2^n)$, giving the bounded formula.

Proof architecture. There are two nested finite differences. Inside a block of length 2^k , the generating polynomial factors as

$$\sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} z^r = \prod_{j=0}^{k-1} (1 - z^{2^j}). \quad (113)$$

It has a zero of order k at $z = 1$, so the inner power sum annihilates every polynomial of degree below k . After this forced zero is removed, its degree- n coefficient is a polynomial of degree at most n in the geometric scale node -2^k .

Across the scales, the weights

$$w_{n,k} = (-1)^k 2^{-\binom{k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_{1/2} \quad (114)$$

form the divided-difference stencil on the nodes $-1, -2, \dots, -2^n$. The finite q -binomial theorem says that this stencil annihilates degrees below n and extracts the leading coefficient in degree n . Applying it after the inner Thue–Morse cancellation produces exactly the moment normalization equal to $\mathcal{F}(m/2^n)$.

For translation invariance, package the entire numerator as a polynomial $N_{m,n}(X)$ over $\mathbb{Q}$. Multiplying every block exponential by e^{QX} cannot change its degree- n coefficient, because the aggregated coefficients of all lower degrees have vanished. Thus $N_{m,n}$ is a constant polynomial. Evaluation in any characteristic-zero field proves the rational, real, and complex statements algebraically; no density or continuity argument is involved. Decomposing the long prefix into m blocks of length 2^k supplies the arbitrary numerator m . $\square$

The endpoint cases are literal finite identities. When $m = 0$, every inner range is empty, so both sides are zero. When $n = 0$, the formula is the scale-zero signed block identity. A nonreduced presentation $m/2^n = 2m/2^{n+1}$ changes every finite sum but not its value, because both are identified with the same global function value.

Binary complementation gives a useful raw-coordinate version. Reflection $r \mapsto 2^k - 1 - r$ changes the Thue–Morse sign by $(-1)^k$. Once the exponent $n + k$ is included, the remaining sign is independent of k and is absorbed by $(-2)^{n^2}$. Consequently, for every $n \in \mathbb{N}$ and $Q \in \mathbb{C}$,

$$\mathcal{F}(2^{-n}) = \frac{1}{(-2)^{n^2}(1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} (r+Q)^{n+k}. \quad (115)$$

Again the left side is embedded in $\mathbb{C}$, and the nested expression is independent of Q .

The q -binomial formula is not competitive with the bit recursion as an evaluator. Its value is structural: the inner difference is adapted to binary blocks, the outer Gaussian difference is adapted to geometric scales, and the same identity survives arbitrary complex scalar translation.

10 The global binary-reduction telescope

The Taylor-block recursion also applies to the signed global extension $\mathcal{F}$. It can therefore be iterated along the binary expansion of an arbitrary nonnegative real number, rather than only along the finite expansion of a dyadic rational. The reductions need not terminate, but their residuals tend geometrically to zero.

Define the finite reduction polynomial

$$\mathcal{T}_m(y) = \sum_{j=0}^m 2^{\binom{j+1}{2} - \binom{m-j}{2}} \frac{d_{m-j}}{(m-j)!} \frac{y^j}{j!}. \quad (116)$$

All powers of 2 here are integer powers. In particular, negative exponents are allowed, and $\mathcal{T}_0(y) = 1$. The signed Taylor-block identity is

$$\mathcal{F}(2^{-m} + y) = -\mathcal{F}(y) + \mathcal{T}_m(y), \quad 0 \leq y < 2^{-m}. \quad (117)$$

Fix $x \geq 0$. At scale m , put

$$p_m = \lfloor 2^m x \rfloor, \quad y_m = x - \frac{p_m}{2^m}, \quad 0 \leq y_m < 2^{-m}. \quad (118)$$

The previous prefix requires a genuine half-scale value at $m = 0$:

$$p_{m-1}^* = \begin{cases} \lfloor x/2 \rfloor, & m = 0, \\ p_{m-1}, & m \geq 1. \end{cases} \quad (119)$$

Uniformly, $p_{m-1}^* = \lfloor p_m/2 \rfloor$. Thus $p_m = 2p_{m-1}^* + \delta_m$ with $\delta_m \in \{0, 1\}$. Define

$$\alpha_m(x) = (-1)^{\text{wt}_2(p_m)} (2p_{m-1}^* - p_m), \quad R_m(x) = (-1)^{\text{wt}_2(p_m)} \mathcal{F}(y_m). \quad (120)$$

The coefficient α_m is zero or a sign, and it is nonzero precisely when the newly exposed binary digit is 1.

Lemma 10.1 (One-step residual identity). *For $m \geq 1$,*

$$\alpha_m(x) \mathcal{T}_m(y_m) = R_{m-1}(x) - R_m(x). \quad (121)$$

Proof. If $p_m = 2p_{m-1}$, then $y_m = y_{m-1}$, the Thue–Morse signs agree, and $\alpha_m = 0$, so both sides vanish. If $p_m = 2p_{m-1} + 1$, then

$$y_{m-1} = 2^{-m} + y_m, \quad (-1)^{\text{wt}_2(p_m)} = -(-1)^{\text{wt}_2(p_{m-1})}, \quad \alpha_m = (-1)^{\text{wt}_2(p_{m-1})}.$$

Apply (117) to y_{m-1} and multiply by the old prefix sign; the two copies of $\mathcal{F}(y_m)$ combine exactly into $R_{m-1} - R_m$. $\square$

The scale-zero identity is separate and retains the parity of the integer block:

$$\mathcal{F}(x) = \alpha_0(x)\mathcal{T}_0(y_0) + R_0(x). \quad (122)$$

Indeed, write $p_0 = 2p_{-1}^* + \delta_0$ and use the two-unit translation law (12). When $\delta_0 = 0$, the polynomial term vanishes. When $\delta_0 = 1$, the identity $\mathcal{F}(1 + y_0) = -\mathcal{F}(y_0) + 1$ supplies it.

Theorem 10.2 (Exact finite binary telescope). *For $x \geq 0$ and $N \in \mathbb{N}$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^N \alpha_m(x)\mathcal{T}_m(y_m(x)) + R_N(x).} \quad (123)$$

Proof. Start with (122). Summing (121) for $m = 1, \dots, N$ telescopes all intermediate residuals. The case $N = 0$ is exactly the scale-zero identity. $\square$

For $N \geq 1$, $0 \leq y_N < 2^{-N} \leq 1/2$. On this interval $0 \leq \mathcal{F}(y) = F(y) \leq 2y$, so

$$|R_N(x)| \leq 2^{1-N}. \quad (124)$$

The one-step identity then gives, for $m \geq 2$,

$$|\alpha_m(x)\mathcal{T}_m(y_m)| \leq |R_{m-1}| + |R_m| \leq 4 \cdot 2^{-(m-1)}. \quad (125)$$

Thus the limiting series is absolutely, not merely conditionally, convergent.

Theorem 10.3 (Absolutely convergent global binary series). *For every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x)\mathcal{T}_m(y_m(x)).} \quad (126)$$

On $0 \leq x \leq 1$, the sum is equivalently $F(x)$.

The index $m = 0$ is indispensable globally. For $0 \leq x < 1$ it vanishes, so a series starting at $m = 1$ happens to be valid there. At $x = 1$, however, the zeroth term is 1 and all later coefficients vanish. More generally it is the zeroth term that distinguishes even and odd integer blocks of the signed extension.

10.1 The corrected parity-power presentation

The same summands have a more literal presentation in terms of prefix parity and inverse dyadic values. Define, using integer exponents,

$$\kappa_j = \frac{((j : \mathbb{Z}) + 2)((j : \mathbb{Z}) - 1)}{2} = \binom{j+1}{2} - 1. \quad (127)$$

The integer interpretation matters: $\kappa_0 = -1$, hence $2^{\kappa_0} = 1/2$; natural subtraction would silently give the wrong term. Put

$$\mathcal{I}_m(y) = \sum_{j=0}^m 2^{\kappa_j} \mathcal{F}(2^{(j:\mathbb{Z})-(m:\mathbb{Z})}) \frac{y^j}{j!}. \quad (128)$$

Lemma 10.4. *For every m and y ,*

$$\mathcal{I}_m(y) = \frac{1}{2} \mathcal{T}_m(y). \quad (129)$$

Proof. For $j \leq m$, set $r = m - j$. The inverse-dyadic identity (78) gives

$$\mathcal{F}(2^{j-m}) = \frac{d_{m-j}}{(m-j)! 2^{\binom{m-j}{2}}}.$$

Together with $\kappa_j = \binom{j+1}{2} - 1$, this identifies each summand with one half of the corresponding summand in (116). $\square$

Let $\tau(a) = \text{wt}_2(a) \bmod 2$. Since

$$\frac{(-1)^a - 1}{2} = 2 \left\lfloor \frac{a}{2} \right\rfloor - a,$$

the literal summand

$$\begin{aligned} \Lambda_m(x) &:= ((-1)^{\lfloor 2^m x \rfloor} - 1)(-1)^{\tau(\lfloor 2^m x \rfloor)} \\ &\quad \times \sum_{j=0}^m 2^{((j+2)(j-1))/2} \mathcal{F}(2^{j-m}) \frac{(x - \lfloor 2^m x \rfloor / 2^m)^j}{j!} \end{aligned} \quad (130)$$

is exactly $\alpha_m(x) \mathcal{T}_m(y_m)$. Both exponents $((j+2)(j-1))/2$ and $j-m$ in this display are integer exponents, not truncated natural subtractions.

Theorem 10.5 (Corrected parity-power series). *For every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \Lambda_m(x)}, \quad (131)$$

and the series converges absolutely. On the half-open unit interval $0 \leq x < 1$, the zeroth summand vanishes and one may start at $m = 1$.

Proof. The elementary parity identity and (129) prove $\Lambda_m = \alpha_m \mathcal{T}_m(y_m)$ term by term. The claim follows from (126). The half-open corollary follows from $\lfloor x \rfloor = 0$; the closed endpoint $x = 1$ requires the restored zeroth term. $\square$

10.2 The global complex q -binomial series

The finite constant-polynomial identity from section 9.7 can be inserted at every scale of the binary telescope. For $Q \in \mathbb{C}$, define

$$\mathcal{N}_n^{(Q)} = \sum_{k=0}^n \frac{\lfloor \frac{n}{k} \rfloor 1/2}{4^{\binom{k}{2}} (n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} (r - 2^k + Q)^{n+k}. \quad (132)$$

Translation invariance and the inverse-dyadic normalization give

$$\frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}} (1/2; 1/2)_n} = \frac{2^n d_n}{n!}. \quad (133)$$

The normalization differs from the 2^{n^2} in (112): here it produces a Taylor coefficient rather than a function value.

For $m \in \mathbb{N}$, put

$$\mathcal{Q}_m^{(Q)}(y) = 2^{-\binom{m+1}{2}} \sum_{n=0}^m \frac{(2^{m+1}y)^{m-n}}{(m-n)!} \frac{\mathcal{N}_n^{(Q)}}{2^{\binom{n}{2}} (1/2; 1/2)_n}. \quad (134)$$

Theorem 10.6 (q -binomial Taylor identity). *For every m , every y , and every $Q \in \mathbb{C}$,*

$$\mathcal{Q}_m^{(Q)}(y) = \mathcal{T}_m(y), \quad (135)$$

where the real polynomial on the right is embedded in $\mathbb{C}$. In particular, each finite summand is pointwise independent of Q before any infinite sum is taken.

Proof. Insert (133) into (134) and set $j = m - n$. Collecting the powers of 2 gives

$$2^{-(\binom{m+1}{2})} 2^{(m+1)j} 2^{m-j} = 2^{(\binom{j+1}{2}) - (\binom{m-j}{2})}.$$

The remaining factorials and moments are exactly those in (116). $\square$

Theorem 10.7 (Global complex q -binomial series). *For fixed $Q \in \mathbb{C}$ and every $x \geq 0$,*

$$\boxed{\mathcal{F}(x) = \sum_{m=0}^{\infty} \alpha_m(x) \mathcal{Q}_m^{(Q)}(y_m(x))}, \quad (136)$$

with the left side embedded in $\mathbb{C}$. The series converges absolutely.

This is not a new infinite basic-hypergeometric identity. The outer sum is the analytic binary telescope, while at each scale a finite q -binomial identity presents the same Taylor polynomial. Expanding one summand makes the architecture explicit:

$$\begin{aligned} & \alpha_m(x) 2^{-(\binom{m+1}{2})} \sum_{n=0}^m \frac{(2^{m+1}x - 2p_m(x))^{m-n}}{(m-n)!} \frac{1}{2^{\binom{n}{2}}(1/2; 1/2)_n} \\ & \times \sum_{k=0}^n \frac{[n]_k]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \sum_{r=0}^{2^k-1} (-1)^{\text{wt}_2(r)} (r - 2^k + Q)^{n+k}. \end{aligned} \quad (137)$$

Convergence comes entirely from the binary-tail majorant (125); the special-function manipulations are finite and algebraic.

11 A complex q -binomial discrete limit

There is a second way in which the same Gaussian row produces $\mathcal{F}$. It is a genuine limit of finite shifted splines rather than a sequence of partial sums of (136). Reuse the safe midpoint cutoff $N_p(x)$ from (38). For $x \geq 0$ it is equivalent to the inclusive source range

$$0 \leq r \leq \left\lfloor 2^p x - \frac{1}{2} \right\rfloor, \quad (138)$$

including the empty case where the upper endpoint is -1 .

For fixed $Q \in \mathbb{C}$, define

$$\begin{aligned} \mathcal{D}_n^{(Q)}(x) &= \frac{1}{2^{n^2}(1/2; 1/2)_n} \sum_{k=0}^n \frac{[n]_k]_{1/2}}{4^{\binom{k}{2}}(n+k)!} \\ & \times \sum_{r=0}^{N_{n+k}(x)-1} (-1)^{\text{wt}_2(r)} (r - 2^{n+k}x + Q)^{n+k}. \end{aligned} \quad (139)$$

Theorem 11.1 (Complex discrete-limit formula). *For every fixed $Q \in \mathbb{C}$ and $x \geq 0$,*

$$\boxed{\mathcal{D}_n^{(Q)}(x) \longrightarrow \mathcal{F}(x)} \quad (140)$$

in $\mathbb{C}$. If $0 \leq x \leq 1$, the limit is equivalently $F(x)$.

Proof. For a degree p and complex shift Q , introduce the finite shifted spline

$$S_p^{(Q)}(x) = \frac{(-1)^p}{2^{\binom{p}{2}} p!} \sum_{r=0}^{N_p(x)-1} (-1)^{\text{wt}_2(r)} (r - 2^p x + Q)^p. \quad (141)$$

At $Q = 1/2$ this is exactly the centered spline (39), so by (46),

$$S_p^{(1/2)}(x) \longrightarrow \mathcal{F}(x).$$

Expanding a fixed cutoff branch about $Q = 1/2$ gives an exact finite Taylor combination of lower-degree centered branches. The normalized Thue–Morse bounds then give, for $p \geq 1$,

$$\left| S_p^{(Q)}(x) - S_p^{(1/2)}(x) \right| \leq 2^{-(p-1)} \exp\left(\left|Q - \frac{1}{2}\right|\right). \quad (142)$$

Thus every fixed complex shift has the same spline limit.

Now reindex the outer sum in (139) by $j = n - k$. Direct collection of signs, powers, and factorials gives the exact finite identity

$$\mathcal{D}_n^{(Q)}(x) = \sum_{j=0}^n \omega_{n,j} S_{2n-j}^{(Q)}(x), \quad (143)$$

where

$$\omega_{n,j} = \frac{(-1)^j \left[\begin{smallmatrix} n \\ j \end{smallmatrix} \right]_{1/2} 2^{-(j+1)}}{(1/2; 1/2)_n}. \quad (144)$$

The finite q -binomial theorem gives the row normalization

$$\sum_{j=0}^n \omega_{n,j} = 1. \quad (145)$$

Applying it with the opposite sign also gives the uniform variation bound

$$\sum_{j=0}^n |\omega_{n,j}| = \frac{(-1/2; 1/2)_n}{(1/2; 1/2)_n} \leq 16. \quad (146)$$

Finally, every sampled degree satisfies $2n - j \geq n$. A finite-row Toeplitz lemma therefore carries the convergence of $S_p^{(Q)}(x)$ to the weighted rows (143), proving (140). $\square$

Two distinctions prevent common misreadings of this theorem. First, unlike the complete finite dyadic numerator and each summand of the global q -series, a finite value $\mathcal{D}_n^{(Q)}(x)$ can depend on Q ; it is the limit that is independent of a fixed shift. Second, the Toeplitz row (143) is not a finite binary telescope or a partial sum of (136). These finite expressions share a limit but are not equal term by term.

12 Endpoint asymptotics: exact logarithmic decomposition

We now turn to the behavior of $F(x)$ as $x \downarrow 0$. This is the technically deepest part of the theory and the place where a very small, but mathematically unavoidable, log-periodic fluctuation appears.

Throughout the asymptotic sections, put

$$L = \log 2. \quad (147)$$

We use the standard Stieltjes convention

$$\zeta(1+s) = \frac{1}{s} + \gamma - \gamma_1 s + \mathcal{O}(s^2), \quad (148)$$

where γ is Euler's constant and γ_1 is the first Stieltjes constant.

12.1 The negative Laplace product

The function G from (69) is the moment generating function of the random variable X in (26):

$$G(z) = \mathbb{E}e^{zX}. \quad (149)$$

Indeed, this also follows from (71) and $Y = 2X - 1$. Therefore, for $s > 0$, $G(-s)$ is the ordinary Laplace transform of the probability law.

Iterating the functional equation (70) in the negative direction gives an exact product.

Proposition 12.1 (Negative Laplace product). *For $s > 0$,*

$$G(-s) = \prod_{j=1}^{\infty} \frac{1 - e^{-s/2^j}}{s/2^j}. \quad (150)$$

Consequently

$$\Lambda(s) := \log G(-s) = \sum_{j=1}^{\infty} \kappa(s/2^j), \quad \kappa(u) = \log \frac{1 - e^{-u}}{u}. \quad (151)$$

The series converges absolutely and locally uniformly on $(0, \infty)$.

Proof. Putting $z = -s/2$ in (70) gives

$$G(-s) = \frac{1 - e^{-s/2}}{s/2} G(-s/2).$$

After N iterations, the remaining factor is $G(-s/2^N)$, which tends to $G(0) = 1$. For small u ,

$$\kappa(u) = -\frac{u}{2} + \mathcal{O}(u^2),$$

so the logarithmic series is dominated on compact s -intervals by a geometric series. $\square$

The product has the exact dilation law

$$\Lambda(2s) = \kappa(s) + \Lambda(s). \quad (152)$$

This equation resembles a renewal equation on the logarithmic scale. Its homogeneous solutions are periodic functions of $\log_2 s$, which is the conceptual origin of the fluctuation in the final asymptotic.

12.2 Removing the quadratic drift

Define the forward tail

$$T(s) = \sum_{m=0}^{\infty} \log(1 - e^{-s2^m}) < 0 \quad (153)$$

and the positive tail error

$$E(s) = -T(s) \geq 0. \quad (154)$$

The series is exponentially small as $s \rightarrow \infty$.

Lemma 12.2 (Forward-tail bound). *For $s > 0$,*

$$0 \leq E(s) \leq \frac{e^{-s}}{(1 - e^{-s})^2}. \quad (155)$$

In particular $E(s) \leq 4e^{-s}$ for $s \geq \log 2$.

Proof. For $0 < q < 1$, $-\log(1 - q) \leq q/(1 - q)$. Hence

$$E(s) \leq \frac{1}{1 - e^{-s}} \sum_{m \geq 0} e^{-s2^m}.$$

Since $2^m \geq m + 1$, the last sum is at most $\sum_{m \geq 0} e^{-s(m+1)} = e^{-s}/(1 - e^{-s})$. $\square$

For $t \in \mathbb{R}$, set

$$P(t) = \Lambda(2^t) + \frac{L}{2}(t^2 - t) + T(2^t). \quad (156)$$

Theorem 12.3 (Exact periodic decomposition). *The function P is 1-periodic and smooth. For every $s > 0$, with $t = \log_2 s$, one has the exact identity*

$$\Lambda(s) = -\frac{(\log s)^2}{2L} + \frac{\log s}{2} + P(\log_2 s) + E(s). \quad (157)$$

Proof. Write $s = 2^t$. Using (152),

$$\Lambda(2s) = \Lambda(s) + \log(1 - e^{-s}) - \log s.$$

Also

$$T(2s) = T(s) - \log(1 - e^{-s}).$$

The change in the quadratic correction is

$$\frac{L}{2}((t+1)^2 - (t+1) - (t^2 - t)) = Lt = \log s.$$

The three changes cancel, proving $P(t+1) = P(t)$. Rearranging (156) and using $E = -T$ yields (157). Smoothness follows by differentiating locally uniformly convergent series; periodicity then makes every derivative bounded. $\square$

Let

$$\bar{P} = \int_0^1 P(t) dt, \quad \Psi(t) = P(t) - \bar{P}. \quad (158)$$

Then Ψ is smooth, real-valued, 1-periodic, and has mean zero.

12.3 Fourier coefficients of the periodic correction

The periodic term is not an unspecified bounded nuisance: all of its Fourier coefficients are explicit. Put

$$\chi_k = \frac{2\pi i k}{L} \quad (k \in \mathbb{Z}), \quad (159)$$

and use the convention

$$\widehat{\Psi}(k) = \int_0^1 \Psi(t) e^{-2\pi i k t} dt. \quad (160)$$

Theorem 12.4 (Gamma-zeta Fourier coefficients). *One has $\widehat{\Psi}(0) = 0$, and for $k \neq 0$,*

$$\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k) \zeta(1 - \chi_k)}{L}. \quad (161)$$

Every nonzero Fourier coefficient is nonzero. In particular, Ψ is genuinely nonconstant.

Proof. The zero coefficient is zero by definition. For $k \neq 0$, change variables $s = 2^t$ in (160). Dyadic periodization turns the integral over one logarithmic period into a Mellin transform. The basic Mellin identity is

$$\int_0^\infty s^{z-1} \log(1 - e^{-s}) ds = -\Gamma(z)\zeta(1+z), \quad \Re z > 0. \quad (162)$$

It follows by expanding $\log(1 - e^{-s}) = -\sum_{m \geq 1} e^{-ms}/m$ and integrating term by term. Near $s = 0$ the kernel contains logarithmic and linear divergences; the quadratic correction in (156) is exactly the finite-part subtraction that removes them. Analytic continuation of the regularized Mellin transform to the imaginary axis gives (161) at $z = -\chi_k$.

The Gamma function has no zeros. Moreover $1 - \chi_k$ lies on the line $\Re z = 1$ away from the pole at 1, and the zeta function has no zeros there. Hence the product in (161) is nonzero. $\square$

Conjugate symmetry $\widehat{\Psi}(-k) = \overline{\widehat{\Psi}(k)}$ reconstructs a real-valued Fourier series. Stirling's estimate for Γ on vertical lines implies exponential decay of the coefficients, so the Fourier series and all of its differentiated series converge rapidly.

The first coefficient is tiny:

$$\widehat{\Psi}(1) \approx (7.5586467408 - 7.4700963646i) \times 10^{-7}, \quad (163)$$

with magnitude approximately $1.0627110626 \times 10^{-6}$. Thus the visible oscillation is on the scale of a few parts in a million in the logarithm. Tiny is not the same as zero: its nonvanishing changes the logical status of an asymptotic equivalent.

12.4 The finite part at the Mellin origin

The constant that enters the saddle asymptotic comes from the finite part of $-\Gamma(z)\zeta(1+z)$ at $z = 0$. Using

$$\begin{aligned} \Gamma(z) &= \frac{1}{z} - \gamma + \left(\frac{\gamma^2}{2} + \frac{\pi^2}{12} \right) z + \mathcal{O}(z^2), \\ \zeta(1+z) &= \frac{1}{z} + \gamma - \gamma_1 z + \mathcal{O}(z^2), \end{aligned}$$

one finds

$$-\Gamma(z)\zeta(1+z) + \frac{1}{z^2} = A_0 + \mathcal{O}(z), \quad (164)$$

where

$$A_0 = \frac{6\gamma^2 + 12\gamma_1 - \pi^2}{12}. \quad (165)$$

The final constant is

$$C_F = \frac{A_0}{L} - \frac{7L}{12} - \frac{1}{2} \log \pi \approx -2.02798389863885942397. \quad (166)$$

The terms $-7L/12$ and $-(1/2) \log \pi$ arise from the dyadic Euler–Maclaurin finite part and the leading Gaussian saddle mass, respectively.

13 The corrected sharp asymptotic

13.1 The lower Lambert coordinate

For sufficiently small $x > 0$, define

$$W = W_{-1}(-Lx), \quad (167)$$

where W_{-1} is the lower real branch of Lambert's W -function, and put

$$\lambda = \lambda(x) = -\frac{W}{L}. \quad (168)$$

Then $\lambda \rightarrow \infty$ as $x \downarrow 0$ and

$$\lambda 2^{-\lambda} = x. \quad (169)$$

Equivalently,

$$r = 2^\lambda = \frac{\lambda}{x}. \quad (170)$$

The real number r is the explicit Lambert saddle radius used in the contour analysis.

13.2 The main term

Define

$$\mathcal{M}(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log \lambda + C_F + \Psi(\lambda), \quad (171)$$

where C_F is given by (166) and $\lambda = \lambda(x)$. The same expression can be written compactly in the Lambert variable W :

$$\mathcal{M}(x) = -\frac{7L}{12} - \frac{1}{2}\log(\pi x) + \frac{A_0 - W - \frac{1}{2}W^2}{L} + \Psi\left(-\frac{W}{L}\right). \quad (172)$$

The equivalence follows from $W = -L\lambda$ and $\log x = \log \lambda - L\lambda$.

Theorem 13.1 (Corrected sharp endpoint asymptotic). *As $x \downarrow 0$,*

$$\boxed{\log F(x) = \mathcal{M}(x) + \mathcal{O}\left(\frac{1}{\lambda(x)}\right) = \mathcal{M}(x) + \mathcal{O}\left(\frac{1}{-\log x}\right)}. \quad (173)$$

Equivalently,

$$F(x) \sim \exp \mathcal{M}(x). \quad (174)$$

The periodic term is evaluated at the *exact* phase $\lambda(x)$, not at a truncated log-log approximation to it. This matters at the stated error scale.

13.3 Bromwich inversion

Because $G(-z) = \mathbb{E}e^{-zX}$ is entire and F is continuous, Laplace–Stieltjes inversion gives, for every $r > 0$ and $0 < x < 1$,

$$F(x) = \frac{1}{2\pi i} \int_{r-i\infty}^{r+i\infty} \frac{e^{zx} G(-z)}{z} dz. \quad (175)$$

Set $z = r(1 + i\theta)$ and choose $r = 2^\lambda = \lambda/x$. Then

$$F(x) = \frac{e^{\lambda + \Lambda(r)}}{2\pi} \int_{-\infty}^{\infty} e^{i\lambda\theta} \frac{G(-r(1 + i\theta))}{G(-r)} \frac{d\theta}{1 + i\theta}. \quad (176)$$

The exact real decomposition (157) at $r = 2^\lambda$ gives

$$\lambda + \Lambda(r) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda + P(\lambda) + E(r). \quad (177)$$

Since

$$\bar{P} = \frac{A_0}{L} - \frac{L}{12}, \quad \bar{P} - \frac{1}{2}\log(2\pi) = C_F, \quad (178)$$

the missing factors in (171) are exactly the leading Gaussian mass $1/\sqrt{2\pi\lambda}$ and the centered fluctuation $\Psi = P - \bar{P}$.

13.4 The local phase

Let

$$u = \log(1 + i\theta), \quad (179)$$

with the principal logarithm near the saddle. Subtracting the value at $\theta = 0$ from the complexified version of (157), and including the factor $(1 + i\theta)^{-1}$ in (176), gives the local phase

$$\Phi_\lambda(\theta) = \lambda(i\theta - u) - \frac{u}{2} - \frac{u^2}{2L} + \Psi\left(\lambda + \frac{u}{L}\right) - \Psi(\lambda) + \mathcal{E}_r(\theta), \quad (180)$$

where $\mathcal{E}_r(0) = 0$ and, on a fixed neighborhood of zero, every derivative of $\mathcal{E}_r$ is exponentially small in r . This last term comes from the forward tail E and is far smaller than every power of $1/\lambda$.

The elementary expansion

$$\log(1 + i\theta) = i\theta + \frac{\theta^2}{2} - \frac{i\theta^3}{3} - \frac{\theta^4}{4} + \mathcal{O}(\theta^5) \quad (181)$$

shows that

$$\lambda(i\theta - u) = -\frac{\lambda\theta^2}{2} + \frac{i\lambda\theta^3}{3} + \frac{\lambda\theta^4}{4} + \mathcal{O}(\lambda|\theta|^5). \quad (182)$$

Thus the natural central scale is

$$\theta = \frac{v}{\sqrt{\lambda}}. \quad (183)$$

Uniformly for $|v|$ growing sufficiently slowly with λ ,

$$\Phi_\lambda(v/\sqrt{\lambda}) = -\frac{v^2}{2} + \lambda^{-1/2}P_1(v; \lambda) + \lambda^{-1}P_2(v; \lambda) + \mathcal{O}\left(\lambda^{-3/2}(1 + |v|^9)\right), \quad (184)$$

where

$$A(u) = \frac{\Psi'(u)}{L} - \frac{1}{2}, \quad (185)$$

$$B(u) = -\frac{1}{4} + \frac{1}{2L} + \frac{\Psi'(u)}{2L} - \frac{\Psi''(u)}{2L^2}, \quad (186)$$

$$P_1(v; u) = i \left(A(u)v + \frac{v^3}{3} \right), \quad (187)$$

$$P_2(v; u) = \frac{v^4}{4} + B(u)v^2. \quad (188)$$

In (184), the second argument is $u = \lambda$.

13.5 Central and tail estimates

For completeness, the quantitative saddle proof is organized by the following estimates. They are stated here in the form needed for the argument; the formal development proves them with explicit domains and constants.

Lemma 13.2 (Central expansion). *For every fixed truncation order N , there is $\delta > 0$ such that on $|\theta| \leq \delta$ the exponent in (176) admits a Taylor expansion through order N after the scaling (183). Its coefficients are smooth 1-periodic functions of λ , and the remainder is dominated by a polynomial in v times the next power of $\lambda^{-1/2}$.*

Proof. The logarithm in (179) is analytic for $|\theta| < 1$. The periodic function Ψ is smooth with bounded derivatives of every order. Taylor's theorem applied to each term of (180), together with the exponentially small bounds on $\mathcal{E}_r$, gives a uniform expansion. The dependence on λ occurs only through $\Psi^{(j)}(\lambda)$, hence is periodic. $\square$

Lemma 13.3 (Gaussian domination). *There are positive constants c, δ and λ_0 such that, for $\lambda \geq \lambda_0$ and $|\theta| \leq \delta$,*

$$\Re \Phi_\lambda(\theta) \leq -c\lambda\theta^2. \quad (189)$$

Proof architecture. The exact function $\Re(i\theta - \log(1 + i\theta)) = -(1/2)\log(1 + \theta^2)$ supplies the negative quadratic term. All remaining terms in (180) have bounded first and second derivatives on a fixed neighborhood. For large λ the $-\lambda\theta^2/2$ contribution dominates them. $\square$

Lemma 13.4 (Off-saddle decay). *For each N one may choose the central window so that the contribution to (176) outside that window is $O(\lambda^{-N})$ relative to the leading saddle mass.*

Proof architecture. Split the negative-Laplace product at the dyadic index nearest λ . In the intermediate range, a positive proportion of the factors have arguments of order one and their moduli are strictly smaller off the real axis, yielding exponential decay in $\lambda\theta^2$. In the far range, repeated use of the sinc/Laplace product bounds gives arbitrarily high polynomial decay. The two bounds overlap and cover the whole complement of the central window. The factor $(1 + i\theta)^{-1}$ is harmless. $\square$

These lemmas justify extending the scaled central integral to the entire real v -axis and integrating its expansion term by term against $e^{-v^2/2}$.

13.6 Completion of the leading proof

Keeping only the Gaussian term in (184) gives

$$\int_{\mathbb{R}} e^{\Phi_\lambda(\theta)} d\theta = \frac{1}{\sqrt{\lambda}} \int_{\mathbb{R}} e^{-v^2/2} dv \left(1 + \mathcal{O}(\lambda^{-1})\right) = \sqrt{\frac{2\pi}{\lambda}} \left(1 + \mathcal{O}(\lambda^{-1})\right). \quad (190)$$

The term of order $\lambda^{-1/2}$ integrates to zero because P_1 is odd in v and purely imaginary. Combining (177), (178), and (190) yields

$$\log F(x) = -\frac{L}{2}\lambda^2 + \left(1 + \frac{L}{2}\right)\lambda - \frac{1}{2}\log \lambda + C_F + \Psi(\lambda) + \mathcal{O}(\lambda^{-1}),$$

which is (173). Finally, $\lambda \asymp -\log x$ follows from (169), so the two displayed error scales are equivalent.

14 First correction and the all-orders saddle expansion

The leading theorem leaves an error of order $1/\lambda$. The same contour analysis can be continued to arbitrary order. The coefficient functions are periodic because every local coefficient is built from derivatives of Ψ evaluated at the exact phase λ .

14.1 The first correction

Expand the exponential in (184):

$$e^{\Phi_\lambda(v/\sqrt{\lambda})} = e^{-v^2/2} \left[1 + \lambda^{-1/2} P_1(v; \lambda) \right. \quad (191)$$

$$\left. + \lambda^{-1} \left(P_2(v; \lambda) + \frac{1}{2} P_1(v; \lambda)^2 \right) + \mathcal{O}(\lambda^{-3/2}(1 + |v|^{12})) \right]. \quad (192)$$

Let Z be a standard normal random variable. Since

$$\mathbb{E}Z^2 = 1, \quad \mathbb{E}Z^4 = 3, \quad \mathbb{E}Z^6 = 15, \quad (193)$$

the normalized coefficient of λ^{-1} in the saddle mass is

$$\begin{aligned} b_1(u) &= \mathbb{E} \left[P_2(Z; u) + \frac{1}{2} P_1(Z; u)^2 \right] \\ &= \frac{3}{4} + B(u) - \frac{1}{2} \left(A(u)^2 + 2A(u) + \frac{5}{3} \right). \end{aligned} \quad (194)$$

Substituting (185) and (186) simplifies the expression considerably.

Theorem 14.1 (First saddle correction). *Define*

$$A_1(u) = \frac{1}{24} + \frac{1}{2L} - \frac{\Psi''(u) + \Psi'(u)^2}{2L^2}. \quad (195)$$

Then

$$\log F(x) = \mathcal{M}(x) + \frac{A_1(\lambda(x))}{\lambda(x)} + \mathcal{O}(\lambda(x)^{-2}). \quad (196)$$

Proof. The order- $\lambda^{-1/2}$ term in (192) is odd and integrates to zero. Equation (194) gives the first nonzero relative correction to the integral. A direct simplification yields (195). If the normalized saddle mass is $1 + b_1/\lambda + O(\lambda^{-2})$, then its logarithm is $b_1/\lambda + O(\lambda^{-2})$, so the same coefficient appears in the logarithmic expansion. The off-saddle contribution is smaller than the stated remainder by the tail estimates of the preceding section. $\square$

The constant part $1/24 + 1/(2L)$ is the usual type of Gaussian/curvature correction. The terms involving Ψ' and Ψ'' record the fact that the log-periodic environment is not constant across the saddle width.

14.2 All orders

At order N , Taylor expansion of (180) produces finitely many polynomials in v whose coefficients are differential polynomials in

$$\Psi'(u), \Psi''(u), \dots, \Psi^{(2N)}(u).$$

After exponentiation, all odd powers of $\lambda^{-1/2}$ integrate to zero by parity. The remaining Gaussian moments are integers, so one obtains integer powers of $1/\lambda$.

Theorem 14.2 (Full logarithmic expansion). *There are smooth 1-periodic functions $A_j : \mathbb{R} \rightarrow \mathbb{R}$, $j \geq 0$, with $A_0 = 0$ and A_1 given by (195), such that for every $N \geq 1$,*

$$\log F(x) = \mathcal{M}(x) + \sum_{j=1}^{N-1} \frac{A_j(\lambda(x))}{\lambda(x)^j} + \mathcal{O}(\lambda(x)^{-N}) \quad (197)$$

as $x \downarrow 0$.

Proof architecture. Fix N . Expand the analytic local phase through sufficiently high order in $\lambda^{-1/2}$, with a Gaussian-dominated remainder. Exponentiate using complete Bell polynomials, integrate term by term, and discard odd terms. The quantitative tail theorem makes the complement of the central region $O(\lambda^{-N})$. This gives a relative saddle mass expansion

$$1 + \sum_{j=1}^{N-1} \frac{b_j(\lambda)}{\lambda^j} + O(\lambda^{-N}), \quad (198)$$

where every b_j is smooth and periodic. Taking the formal logarithm defines the A_j . If

$$\log \left(1 + \sum_{j \geq 1} b_j z^j \right) = \sum_{j \geq 1} A_j z^j,$$

then the coefficients may be generated recursively by

$$A_n = b_n - \frac{1}{n} \sum_{k=1}^{n-1} k A_k b_{n-k}. \quad (199)$$

All operations preserve smooth periodic dependence on the phase. $\square$

This form of the expansion is preferable to a very long series in nested logarithms. The exact Lambert coordinate absorbs the dominant implicit inversion, while the exact periodic phase prevents phase errors from being magnified into false remainder claims.

15 Elementary log-log forms and the missing periodic term

Lambert W gives the cleanest formula, but an expansion in elementary logarithms is useful for comparison with published displays. It must, however, retain Ψ at the exact Lambert phase.

15.1 The literal small- x expansion

Put

$$S = -\log x, \quad m = \log S, \quad a = \log L. \quad (200)$$

Thus $S \rightarrow \infty$ as $x \downarrow 0$. Define the nonperiodic elementary expression

$$\begin{aligned} \mathcal{E}_0(x) = & -\frac{S^2}{2L} - \frac{Sm}{L} + \left(\frac{1}{2} + \frac{1+a}{L} \right) S \\ & - \frac{m^2}{2L} + \frac{am}{L} + \frac{6\gamma^2 + 12\gamma_1 - \pi^2 - 6a^2}{12L} - \frac{7L}{12} - \frac{1}{2} \log \pi \\ & - \frac{m^2}{2LS} + \frac{am}{LS}. \end{aligned} \quad (201)$$

Theorem 15.1 (Corrected elementary expansion). *As $x \downarrow 0$,*

$$\log F(x) = \mathcal{E}_0(x) + \Psi(\lambda(x)) + \mathcal{O}\left(\frac{1}{-\log x}\right). \quad (202)$$

Proof architecture. The standard lower-branch Lambert expansion solves $\lambda L - \log \lambda = S$ by successive substitution. Substituting that expansion into (171), and retaining enough phase terms to control the amplified quadratic contribution, gives (201). A coarse two-term approximation to λ is insufficient: because the main exponent contains $-L\lambda^2/2$, an error $\delta\lambda$ contributes about $-L\lambda\delta\lambda$, so the phase must be known one order more accurately than a naive count suggests. The fourth Lambert phase term in the formal proof is exactly what reduces the residual to $O(1/S)$. $\square$

In the original variable $\ell = \log x < 0$, the same expression is

$$\begin{aligned} \mathcal{E}_0(x) = & -\frac{\ell^2}{2L} + \frac{\ell \log(-\ell)}{L} - \left(\frac{1}{2} + \frac{1 + \log L}{L}\right) \ell \\ & - \frac{\log^2(-\ell)}{2L} + \frac{\log L \log(-\ell)}{L} \\ & + \frac{6\gamma^2 + 12\gamma_1 - \pi^2 - 6 \log^2 L}{12L} - \frac{7L}{12} - \frac{1}{2} \log \pi \\ & + \frac{\log^2(-\ell)}{2L\ell} - \frac{\log L \log(-\ell)}{L\ell}. \end{aligned} \quad (203)$$

This is the literal elementary display formalized in the repository, with the periodic term added separately.

15.2 The dyadic logarithmic scale

Set $x = 2^{-t}$ and let

$$\lambda_t = -\frac{1}{L} W_{-1}(-L2^{-t}). \quad (204)$$

Then λ_t is the unique large solution of

$$\lambda_t - \log_2 \lambda_t = t. \quad (205)$$

Its first terms are

$$\lambda_t = t + \frac{\log t}{L} + \frac{\log t}{L^2 t} + \frac{\log t - \frac{1}{2} \log^2 t}{L^3 t^2} + \mathcal{O}\left(\frac{\log^3 t}{t^3}\right). \quad (206)$$

Define

$$\mathcal{D}_0(t) = -\frac{L}{2} t^2 - t \log t + \left(1 + \frac{L}{2}\right) t - \frac{\log^2 t}{2L} + C_F - \frac{\log^2 t}{2L^2 t}. \quad (207)$$

Theorem 15.2 (Corrected dyadic-scale expansion). *As $t \rightarrow \infty$,*

$$\boxed{\log F(2^{-t}) = \mathcal{D}_0(t) + \Psi(\lambda_t) + \mathcal{O}(t^{-1})}. \quad (208)$$

The literal substitution of $x = 2^{-t}$ into (201) differs from $\mathcal{D}_0(t)$ by the explicit term

$$\frac{\log^2 L}{2L^2 t}, \quad (209)$$

which is itself $O(1/t)$. Thus both nonperiodic displays are valid at the stated precision, but they are not algebraically identical.

15.3 Why the uncorrected formula is false at the claimed scale

Let $\mathcal{M}_0(x) = \mathcal{M}(x) - \Psi(\lambda(x))$ be the nonperiodic Lambert main. From (173),

$$\log F(x) - \mathcal{M}_0(x) = \Psi(\lambda(x)) + O(1/\lambda(x)). \quad (210)$$

Since Ψ is nonconstant, the right side does not tend to zero.

Proposition 15.3 (Failure without the periodic correction). *The relation*

$$\log F(x) = \mathcal{M}_0(x) + O(1/(-\log x)) \quad (211)$$

is false. In fact $F(x)/\exp(\mathcal{M}_0(x))$ has distinct subsequential limits.

Proof. Choose $u, v \in [0, 1)$ with $\Psi(u) \neq \Psi(v)$. Set

$$\lambda_n = n + u, \quad x_n = \lambda_n 2^{-\lambda_n}.$$

Then $x_n \downarrow 0$ and the exact Lambert phase of x_n is λ_n . Hence

$$\log F(x_n) - \mathcal{M}_0(x_n) \longrightarrow \Psi(u).$$

The analogous sequence with phase $n + v$ tends to $\Psi(v)$. Since $1/(-\log x) \rightarrow 0$, the bound (211) is impossible. Exponentiating gives distinct ratio limits $e^{\Psi(u)}$ and $e^{\Psi(v)}$. $\square$

Numerically the discrepancy is minute because the first Fourier mode has size about 10^{-6} , but asymptotic equivalence is a limit statement, not a graphical tolerance. A nonzero bounded oscillation cannot be hidden inside an error term that tends to zero.

15.4 Symmetry at the right endpoint

The symmetry $F(1 - x) = 1 - F(x)$ immediately transfers the left-endpoint expansion to the right:

$$1 - F(1 - x) = F(x). \quad (212)$$

Thus the same Lambert phase, periodic fluctuation, and all-orders coefficients describe the survival probability near 1.

16 Consequences of the endpoint expansion

The full sharp formula immediately settles several qualitative questions that are otherwise awkward to read from the differential equation.

Theorem 16.1 (Quadratic logarithmic law). *As $x \downarrow 0$,*

$$\frac{\log F(x)}{(\log x)^2} \longrightarrow -\frac{1}{2 \log 2}. \quad (213)$$

Equivalently, on the dyadic scale,

$$\log F(2^{-t}) \sim -\frac{\log 2}{2} t^2. \quad (214)$$

Proof. In (171), $\lambda \sim S/L$ with $S = -\log x$. The quadratic term $-L\lambda^2/2$ dominates the linear term, the logarithmic term, and the bounded periodic correction. Substitution gives (213). $\square$

Corollary 16.2 (Between powers and pure endpoint exponentials). *For every $N > 0$ and every $c > 0$,*

$$F(x) = o(x^N), \quad (215)$$

and

$$e^{-c/x} = o(F(x)) \quad (216)$$

as $x \downarrow 0$.

Proof. The logarithm of x^N is $-NS$, whereas $\log F(x) \sim -S^2/(2L)$, so $F(x)/x^N \rightarrow 0$. On the other hand, $\log e^{-c/x} = -ce^S$, which is much more negative than $-S^2/(2L)$; therefore $e^{-c/x}/F(x) \rightarrow 0$. $\square$

This places the endpoint decay in an intermediate regime. The function is flatter than any finite algebraic order, as every compactly supported smooth function must be at its support boundary, but it is vastly larger than the familiar model $e^{-1/x}$.

Corollary 16.3 (Asymptotics of reciprocal-power values). *For integers $n \rightarrow \infty$,*

$$\begin{aligned} \log F(2^{-n}) = & -\frac{L}{2}n^2 - n \log n + \left(1 + \frac{L}{2}\right)n - \frac{\log^2 n}{2L} + C_F \\ & - \frac{\log^2 n}{2L^2 n} + \Psi(\lambda_n) + O(n^{-1}), \end{aligned} \quad (217)$$

where λ_n is defined by (204). Combining this with (78) gives an equally sharp expansion for the half moments d_n .

The formula shows why a fixed multiplicative constant cannot complete a recurrence-based ansatz for $F(2^{-n})$: the fractional parts of λ_n drift, and the factor $\exp \Psi(\lambda_n)$ is nonconstant.

17 The Fourier–Legendre expansion of up

A second exact spectral representation uses the ordinary Legendre polynomials P_n on $[-1, 1]$. Unlike the Fourier transform, this is a polynomial expansion on the natural support interval, and its coefficients can be expressed by the reciprocal-power dyadic values already computed.

17.1 Orthogonality and coefficients

Normalize the Legendre polynomials by $P_n(1) = 1$. They satisfy

$$-\frac{d}{dx} \left((1-x^2)P'_n(x) \right) = n(n+1)P_n(x), \quad (218)$$

with orthogonality

$$\int_{-1}^1 P_m(x)P_n(x) dx = \frac{2}{2n+1} \delta_{mn}. \quad (219)$$

For a continuous function f , define

$$\alpha_n(f) = \frac{2n+1}{2} \int_{-1}^1 f(x)P_n(x) dx. \quad (220)$$

Because up is even and $P_n(-x) = (-1)^n P_n(x)$,

$$\alpha_{2n+1}(\text{up}) = 0. \quad (221)$$

Put

$$u_n = \alpha_{2n}(\text{up}) = \frac{4n+1}{2} \int_{-1}^1 \text{up}(x)P_{2n}(x) dx. \quad (222)$$

17.2 Exact coefficient formula

The even Legendre polynomial has the finite power expansion

$$P_{2n}(x) = 4^{-n} \sum_{k=0}^n (-1)^{n+k} \binom{2n}{n+k} \binom{2n+2k}{2n} x^{2k}. \quad (223)$$

The moment-to-dyadic bridge is

$$c_k = 2(2k)! 2^{\binom{2k+1}{2}} F(2^{-2k-1}). \quad (224)$$

This is (80) with an even exponent, together with symmetry.

Theorem 17.1 (Exact Legendre coefficients). *For every $n \geq 0$,*

$$u_n = 4^{-n}(4n+1) \sum_{k=0}^n (-1)^{n+k} \binom{2n}{n+k} \binom{2n+2k}{2n} \times (2k)! 2^{\binom{2k+1}{2}} F(2^{-2k-1}). \quad (225)$$

In particular every u_n is rational.

Proof. Insert (223) into (222) and interchange the finite sum with the integral. The integral of $x^{2k} \text{up}(x)$ is c_k . Substitution of (224) cancels the factor $1/2$ in (222) and gives (225). $\square$

Thus the entire polynomial spectral expansion is anchored in the same exact arithmetic table $F(2^{-m})$ that drives the bit-recursive evaluator.

17.3 Absolute and uniform convergence

The Legendre Sturm–Liouville operator

$$\mathcal{L}f = -\frac{d}{dx} \left((1-x^2)f'(x) \right) \quad (226)$$

is formally self-adjoint on $[-1, 1]$. Boundary terms vanish because of the factor $1-x^2$. Repeated integration by parts therefore gives, for $n \geq 1$,

$$\alpha_n(f) = \frac{2n+1}{2[n(n+1)]^m} \int_{-1}^1 (\mathcal{L}^m f)(x) P_n(x) dx. \quad (227)$$

Since $|P_n(x)| \leq 1$ on $[-1, 1]$, any C^∞ function has coefficients that decay faster than every power of n .

Theorem 17.2 (Uniform Legendre series). *On the entire closed interval $[-1, 1]$,*

$$\boxed{\text{up}(x) = \sum_{n=0}^{\infty} u_n P_{2n}(x).} \quad (228)$$

The series converges absolutely and uniformly, including at both endpoints.

Proof. Apply (227) to $f = \text{up}$. Choosing $m \geq 2$ gives $|\alpha_n(\text{up})| \leq Cn^{-3}$, say. Since $|P_n| \leq 1$, the Weierstrass test gives absolute and uniform convergence. Let the uniform sum be g . Orthogonality shows that g and up have the same Legendre coefficients. Completeness of the Legendre polynomials in $L^2[-1, 1]$ implies $g = \text{up}$ almost everywhere, and continuity upgrades equality to every point. Odd coefficients vanish by (221), leaving (228). $\square$

The endpoint statement is worth emphasizing. Generic Fourier–Legendre convergence theorems are often quoted only on the open interval, or with an endpoint average. Here smoothness and absolute uniform convergence give the literal values $\text{up}(\pm 1) = 0$.

17.4 A practical polynomial approximation

Truncating (228) at $n = N$ gives a polynomial of degree $2N$ with exact rational coefficients after expanding the P_{2n} . The error is bounded by

$$\left\| \text{up} - \sum_{n=0}^N u_n P_{2n} \right\|_{\infty} \leq \sum_{n>N} |u_n|. \quad (229)$$

Repeated use of (227) supplies arbitrarily high algebraic tail bounds. For moderate precision this method is attractive: all coefficients are exact, evaluation is stable on $[-1, 1]$, and the approximation respects evenness automatically.

17.5 Least-squares optimality of the partial sums

The Legendre truncation is not only a convenient polynomial approximation; it is the exact orthogonal projection in $L^2[-1, 1]$. Define

$$S_N(x) = \sum_{n=0}^N u_n P_{2n}(x). \quad (230)$$

Theorem 17.3 (Pythagorean error identity). *For every real polynomial p of degree at most $2N + 1$,*

$$\int_{-1}^1 (\text{up}(x) - p(x))^2 dx = \int_{-1}^1 (\text{up}(x) - S_N(x))^2 dx + \int_{-1}^1 (p(x) - S_N(x))^2 dx. \quad (231)$$

Consequently S_N is the unique least-squares approximant among all such polynomials.

Proof. Every polynomial of degree at most $2N + 1$ is a linear combination of $P_0, \dots, P_{2N+1}$. By construction, $\text{up} - S_N$ is orthogonal to each even basis element through P_{2N} ; by evenness it is also orthogonal to every odd basis element. Hence

$$\int_{-1}^1 (\text{up} - S_N)(p - S_N) dx = 0.$$

Expanding $\text{up} - p = (\text{up} - S_N) - (p - S_N)$ gives (231). Equality with the minimum forces the final nonnegative integral to vanish. A polynomial vanishing almost everywhere vanishes identically, so $p = S_N$. $\square$

The one-degree slack is genuine: the odd coefficient of P_{2N+1} is zero. Thus a partial sum visibly of degree $2N$ is already optimal in the larger space of degree at most $2N + 1$.

18 Additional summation and spectral identities

18.1 The cosine series on the support

Periodize up with period 2. Since neighboring support intervals meet only at zeros, the periodization agrees with up on $[-1, 1]$. Its Fourier coefficients are $\widehat{\text{up}}(n/2)/2$, so rapid Fourier decay gives the uniformly convergent cosine series

$$\text{up}(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \widehat{\text{up}}(n/2) \cos(\pi n x), \quad -1 \leq x \leq 1. \quad (232)$$

Using (48), every coefficient is an explicit infinite product. Many coefficients vanish because one of the sinc factors is evaluated at a nonzero integer.

18.2 Scaled partitions

Equation (53) may be differentiated term by term. For $j \geq 1$,

$$\sum_{k \in \mathbb{Z}} \text{up}^{(j)}\left(t + \frac{k}{m}\right) = 0. \quad (233)$$

The sums are locally finite. Combining this with (61) turns the identity into exact finite cancellations among signed translates of $\mathcal{F}$ at dyadic scales.

18.3 Mean and variance of the Fabius law

From the random-series representation,

$$\mathbb{E}X = \sum_{n \geq 0} \frac{1/2}{2^{n+1}} = \frac{1}{2}. \quad (234)$$

Since $\text{Var}(U_n) = 1/12$ and the coordinates are independent,

$$\text{Var}(X) = \frac{1}{12} \sum_{n \geq 0} 4^{-n-1} = \frac{1}{36}. \quad (235)$$

For $Y = 2X - 1$, this gives $\mathbb{E}Y = 0$ and $\text{Var}(Y) = 1/9$, agreeing with $c_1 = 1/9$. Higher even moments are exactly the c_n of (63).

18.4 An exact self-convolution law

Splitting the random series into its even and odd coordinates gives two independent scaled copies with different weights. At the transform level this is the factorization

$$\widehat{\text{up}}(z) = \widehat{\text{up}}(z/2) \text{sinc}(\pi z). \quad (236)$$

In physical space,

$$\text{up}(x) = 2 \int_{-1/2}^{1/2} \text{up}(2x - 2t) dt = \int_{2x-1}^{2x+1} \text{up}(y) dy, \quad (237)$$

with the normalization understood as convolution with the uniform density on $[-1/2, 1/2]$. Differentiating this integral refinement recovers (21). Conversely, integrating the differential equation and using compact support recovers the convolution law.

19 Computation in exact and numerical regimes

The identities above do not lead to a single universally best numerical method. They lead to a collection of algorithms adapted to different questions. This section makes the distinctions explicit, since confusing an exact arithmetic formula with an endpoint asymptotic is as unhelpful as using a dense dyadic table to evaluate one value at 2^{-10^6} .

19.1 Which representation should be used?

Representation	Principal strength	Principal limitation
Contraction fixed point	A direct constructive proof; after m iterations the uniform error is geometrically small.	Produces approximating functions rather than exact individual values.
Finite convolution / probability	Positive approximants, natural sampling, and a transparent explanation of smoothness.	High resolution requires many convolution levels.
Centered finite spline	An explicit finite formula with certified uniform error at most 2^{-p} .	Direct summation has exponentially many terms unless its recursion is compiled.
Fourier sinc product	Entire transform, rapid frequency decay, and efficient evaluation away from extreme endpoint cancellation.	Recovering a tiny value of $F(x)$ by oscillatory inversion may lose many digits.
Exact dyadic formula	A closed rational answer and an algebraic proof of representation invariance.	The literal Thue–Morse sum may contain exponentially many terms.
Bit-recursive Taylor evaluator	Exact rational arithmetic with at most one recursive call per set bit of the numerator.	Rational numerators and denominators themselves become very large.
Legendre series	Uniform polynomial approximation on the full closed support, with exact coefficients.	Not specially adapted to resolving the boundary layer at a tiny x .
Lambert saddle expansion	Stable evaluation of $\log F(x)$ at extraordinarily small positive x .	Asymptotic rather than exact; the periodic correction must be retained.

For the fixed-point construction, let T be the transform of (17). If f_0 is any admissible continuous function, then

$$\|T^m f_0 - F\|_\infty \leq 2^{-m} \|f_0 - F\|_\infty \leq 2^{-m}. \quad (238)$$

Thus the contraction proof itself is a certified numerical scheme. It is excellent for coarse graphs and for constructing a first approximation without any prior arithmetic tables. Exact dyadic computation, however, should use the rational recurrences.

19.2 Precomputing the reciprocal-power table

Put

$$V_n = F(2^{-n}). \quad (239)$$

Eliminating d_n from (72) and (78) gives an especially convenient recurrence involving only earlier V_k :

$$V_0 = 1, \quad V_{n+1} = \frac{1}{2^{n+1} - 1} \sum_{k=0}^n \frac{V_k}{2^{\binom{n+1}{2} - \binom{k}{2}} (n+2-k)!}. \quad (240)$$

Every operation is rational. A practical implementation stores the values in an array, reduces each rational after every addition, and reuses factorials and powers of two. This table is precisely the data required by the Taylor blocks in (100).

There are three useful normalization choices.

1. Store V_n directly. This makes the public API simple and is ideal when values are eventually returned as rational numbers.
2. Store $d_n = n!2^{\binom{n}{2}}V_n$. The recurrence has smaller explicit powers of two, and the parity theorem provides a strong integrity check.
3. Store the division-free natural numerators G_n of the half moments. This avoids fraction reduction inside the recurrence but requires restoring a known structured denominator at the end.

The third choice is usually best for bulk generation; the first is usually best for a small library interface.

19.3 Operation count of the bit recursion

Suppose $0 < a < 2^n$, and write the set-bit positions of a in decreasing order as

$$b_1 > b_2 > \cdots > b_h \geq 0, \quad a = \sum_{j=1}^h 2^{b_j}. \quad (241)$$

The evaluator of [algorithm 9.4](#) makes exactly $h = \text{popcount}(a)$ nontrivial recursive calls. At the j th call, the Taylor order is $n - b_j$. Hence the number of rational multiply-add steps in straightforward Horner evaluation is

$$\sum_{j=1}^h (n - b_j) \leq nh \leq n^2. \quad (242)$$

This is an arithmetic-operation count, not a bit-complexity estimate. The bit lengths of the exact numerator and denominator grow quadratically in n already at $a = 1$, as the factor $2^{\binom{n}{2}}$ in (78) makes unavoidable.

By contrast, the literal formula (94) sums over every $0 \leq h < a$. When a is comparable with 2^n , it therefore has exponentially many outer terms. Its value is conceptual rather than algorithmic: it proves rationality, exposes Thue–Morse cancellation, and connects the answer to q -binomial identities. The Taylor recursion is the compiled form of that cancellation.

Unreduced inputs cause no mathematical difficulty. The exact formula and the executable evaluator satisfy

$$F\left(\frac{2a}{2^{n+1}}\right) = F\left(\frac{a}{2^n}\right). \quad (243)$$

Reducing the pair (a, n) first merely avoids unnecessary work. Signed inputs should be handled at the outermost layer: clamp to 0 for $a \leq 0$, clamp to 1 for $a \geq 2^n$, and call the unit-interval recursion only in the interior.

19.4 Stable endpoint evaluation

For a very small non-dyadic x , direct evaluation of $F(x)$ is the wrong numerical variable. One should compute $\log F(x)$. Set

$$S = -\log x, \quad L = \log 2. \quad (244)$$

The exact saddle coordinate is the positive solution of

$$L\lambda - \log \lambda = S. \quad (245)$$

Although a library implementation may use the W_{-1} branch directly, Newton iteration is simple and stable:

$$\lambda_{j+1} = \lambda_j - \frac{L\lambda_j - \log \lambda_j - S}{L - \lambda_j^{-1}}. \quad (246)$$

For large S , the starting value

$$\lambda_0 = \frac{S + \log S - \log L}{L} \quad (247)$$

already has small relative error. Once λ is known, evaluate (171) and, when needed, add $A_1(\lambda)/\lambda$ from (195).

The periodic function Ψ should be evaluated by its Fourier series (161). The Gamma factor in (161) decays exponentially with the frequency, while the zeta factor has only moderate growth on the relevant vertical line. Consequently a handful of modes usually suffices for precision far beyond that of ordinary floating-point arithmetic. Since Ψ is real,

$$\Psi(u) = \widehat{\Psi}(0) + 2\Re \sum_{k=1}^K \widehat{\Psi}(k)e^{2\pi i k u} + \text{exponentially small Fourier tail}. \quad (248)$$

Reducing u modulo 1 before evaluating the exponentials avoids loss of phase accuracy.

The saddle expansion and the exact dyadic evaluator overlap over a substantial range. Their agreement is a powerful independent test. For $x = 2^{-n}$, an implementation can compare the exact rational logarithm (computed at high precision) against (217). The difference after the first correction should be $O(n^{-2})$, with the phase evaluated at the exact λ_n , not at n .

19.5 A verification suite suggested by the mathematics

An exact implementation should test identities, not merely decimal snapshots. Particularly effective checks are

$$F(a/2^n) + F((2^n - a)/2^n) = 1, \quad (249)$$

$$F(2a/2^{n+1}) = F(a/2^n), \quad (250)$$

$$v_2(F(2^{-n})) = -\binom{n}{2} - 1 - v_2(n!), \quad (251)$$

$$F(5/16) = \frac{305857}{2073600}, \quad (252)$$

$$R_n \equiv 1 \pmod{2}. \quad (253)$$

The Taylor-block identity can be tested for every a on a modest dyadic grid; the q -binomial expression can be evaluated at several unrelated translations q to test its proved translation independence. For floating-point code, the partition of unity (53), Fourier refinement (49), and agreement between the Legendre and Fourier reconstructions provide complementary global tests.

19.6 Effective uniform continuity and sequential computability

Computability has a precise recursion-theoretic meaning here, not merely the informal meaning that a numerical routine can be written. We use the classical Grzegorzczuk formulation [6], expressed through fast dyadic names.

Definition 19.1 (Fast signed-dyadic name). A signed-natural numerator is a pair $c = (c^+, c^-) \in \mathbb{N}^2$. At precision p it represents

$$\langle c \rangle_p = \frac{c^+ - c^-}{2^p}. \quad (254)$$

A real sequence $(x_i)_{i \geq 0}$ is *computable* if one recursive algorithm, uniform in both i and p , produces pairs $a(i, p)$ such that

$$|x_i - \langle a(i, p) \rangle_p| \leq 2^{-p}. \quad (255)$$

A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is *sequentially computable* if it sends every computable real sequence to a computable real sequence.

The two-natural representation is deliberately redundant. It is extensionally the usual signed dyadic representation, but its arithmetic can be certified using primitive recursion on natural numbers without hiding an integer or rational coding convention.

Definition 19.2 (Effective uniform continuity). A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is effectively uniformly continuous if there is a recursive $d: \mathbb{N} \rightarrow \mathbb{N}$ such that, for every positive n ,

$$d(n) > 0, \quad |x - y| < \frac{1}{d(n)} \implies |f(x) - f(y)| < \frac{1}{n}. \quad (256)$$

It is a computable real function in this sense if it is both sequentially computable and effectively uniformly continuous.

Restricting (256) to $n > 0$ and requiring $d(n) > 0$ are substantive, not cosmetic. At $n = 0$ the reciprocal accuracy request is meaningless; in a totalized division formalism, allowing $d(n) = 0$ would make the implication vacuous.

Theorem 19.3 (Computability of the bounded Fabius function). *The constant-tail extension of the bounded Fabius distribution function is sequentially computable and effectively uniformly continuous. A primitive-recursive modulus is*

$$d(n) = 2n \quad (n > 0). \quad (257)$$

Consequently $F: \mathbb{R} \rightarrow [0, 1]$ is a computable real function in the preceding sense.

Proof. The effective-continuity clause follows immediately from the global 2-Lipschitz estimate: if $|x - y| < 1/(2n)$, then

$$|F(x) - F(y)| \leq 2|x - y| < \frac{1}{n}.$$

Multiplication by the constant 2 is primitive recursive, so (257) is an effective modulus.

It remains to give one recursive transformer of fast names. Suppose an output of precision p is requested, and put $s = p + 3$. Read the input numerator $c = (c^+, c^-)$ at precision s , and clamp it to the unit interval by the natural-number operation

$$a = \begin{cases} 0, & c^+ \leq c^-, \\ 2^s, & 2^s + c^- \leq c^+, \\ c^+ - c^-, & \text{otherwise.} \end{cases} \quad (258)$$

Because F has constant tails, this clamp does not change its value. Evaluate the centered spline at the matched grid point $a/2^s$. Formula (39) simplifies to the exact rational number

$$S_s \left(\frac{a}{2^s} \right) = \frac{\sum_{r=0}^{a-1} \varepsilon_r (2a - 2r - 1)^s}{2^{\binom{s+1}{2}} s!}. \quad (259)$$

On this clamped grid the numerator is nonnegative. It can nevertheless be computed without signed arithmetic: keep separate positive and negative accumulators according to the Thue–Morse bit.

Every part of this evaluator is primitive recursive. More explicitly, compute the Thue–Morse bit from

$$b(0) = 0, \quad b(m) = (b(\lfloor m/2 \rfloor) + (m \bmod 2)) \bmod 2 \quad (m > 0), \quad (260)$$

fold the finite range $0 \leq r < a$ using addition, multiplication, subtraction, and powers on natural numbers, and compute the denominator from

$$D_0 = 1, \quad D_{s+1} = D_s 2^{s+1}(s+1), \quad D_s = 2^{\binom{s+1}{2}} s!. \quad (261)$$

Finally round the rational in (259) to the nearest multiple of 2^{-p} , breaking ties upward. If N and D are its nonnegative numerator and positive denominator, the returned numerator is

$$Q = \left\lfloor \frac{2(2^p N) + D}{2D} \right\rfloor. \quad (262)$$

Bounded range folds, natural division, and the displayed recursion are all closed under primitive recursion; thus $(c, p) \mapsto (Q, 0)$ is one certified recursive algorithm.

It remains to verify its precision. The centered-spline error (45) at level $s = p + 3$ costs one unit of 2^{-s} . Nearest-grid rounding costs

$$\frac{1}{2} 2^{-p} = 4 \cdot 2^{-(p+3)}.$$

Hence the evaluator itself has error at most

$$5 \cdot 2^{-(p+3)}. \quad (263)$$

When the input is the s th approximation to an arbitrary real x , propagation through the 2-Lipschitz function costs another

$$2 \cdot 2^{-(p+3)}.$$

The total is therefore

$$(2 + 5)2^{-(p+3)} = 7 \cdot 2^{-(p+3)} \leq 8 \cdot 2^{-(p+3)} = 2^{-p}.$$

Applying the same algorithm uniformly to the two indices (i, p) produces a fast name for $F(x_i)$ from any fast name for (x_i) . This proves sequential computability and completes the theorem. $\square$

Remark 19.4 (Scope of the formal computability theorem). The signed global extension $\mathcal{F}$ is also globally 2-Lipschitz, so the same modulus $d(n) = 2n$ proves its effective uniform continuity. The formal development packages the primitive-recursive spline evaluator and the combined sequential-computability theorem for the bounded constant-tail function F . It does not claim a packaged global sequential computability theorem for $\mathcal{F}$.

20 Corrections, conventions, and proof audit

A formal development is most useful when it records not only what is true but also exactly where an attractive informal shortcut ceases to be valid. The following points are the ones most likely to create a silent error in a human treatment.

20.1 Bounded versus signed-global Fabius functions

The bounded function F takes values in $[0, 1]$ and is constant to the right of 1. The signed extension $\mathcal{F}$ is not bounded in that way and instead obeys

$$\mathcal{F}'(x) = 2\mathcal{F}(2x) \quad (x \in \mathbb{R}).$$

They agree for every $x \leq 1$, hence in particular on $[0, 1]$, but they are globally distinct. Thus a statement such as “ $F'(x) = 2F(2x)$ for all real x ” is correct only when F denotes the signed extension. The explicit Thue–Morse series (11) makes the distinction unavoidable and proves that the global sum is locally finite rather than merely conditionally convergent.

Likewise, the support statement for up concerns the topological support: $\text{supp up} = [-1, 1]$, even though $\text{up}(\pm 1) = 0$. Endpoint formulas must distinguish closed support from the open set on which the function is positive.

20.2 The Reshetnikov normalization

Equation (93) contains the factor $2^{+\binom{n-1}{2}}$. The negative exponent printed in the abstract of one version of the arithmetic paper is incompatible with its own equation (27) and with the integrality theorem. The positive exponent is the one proved by the exact recurrence. The resulting first values are

$$R_1, R_2, \dots, R_7 = 1, 5, 15, 1001, 5985, 2853675, 26261235, \quad (264)$$

all odd as predicted.

The strongest common-denominator formula in the arithmetic paper depends on a separate divisibility conjecture. The formal development keeps the unconditional multiplier bound and the conditional exact denominator statement logically separate. A computational match for the first many levels is evidence for the conjecture, not a proof.

20.3 The local asymptotic derivation

Write the logarithmic endpoint profile in a dyadic variable as

$$g(t) = \log F(2^{-t}). \quad (265)$$

A proposed elementary derivation substitutes a main term $G(t)$ into the exact difference-differential relation. Two steps in that derivation cannot be made for an unspecified remainder $R = g - G$:

1. a coefficient comparison drops the actual difference $R(t) - R(t-1)$;
2. a later estimate uses a bound on $R'(t)$ that has not been assumed or proved.

Even the explicit main term has a residual of order

$$\left(\frac{\log t}{t}\right)^2, \quad (266)$$

with a nonzero leading coefficient; it is not $O(t^{-2})$. Therefore the printed iteration does not establish the claimed bounded periodic remainder. The rigorous route is the exact negative-Laplace decomposition of [section 12](#), followed by Bromwich inversion and a quantitative saddle analysis.

There is also a small but conceptually useful conversion issue. Substituting $x = 2^{-t}$ in the literal elementary small- x expression does not give the separately displayed dyadic expression term for term. Their difference is

$$\frac{(\log \log 2)^2}{2(\log 2)^2 t}, \quad (267)$$

which is $O(t^{-1})$. It is harmless for a coarser theorem but must be retained when the display itself claims accuracy through the inverse-logarithmic term.

20.4 Why the periodic term cannot be averaged away

Replacing $\Psi(\lambda)$ by its mean changes the logarithm by a bounded quantity, so it preserves only very coarse statements such as (213). It does not preserve a sharp multiplicative asymptotic. Indeed, for every nonzero integer k ,

$$\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{\log 2} \neq 0.$$

The nonvanishing follows from the absence of zeros of Γ , the fact that $\zeta(s) \neq 0$ on $\Re s = 1$ away from its pole, and $\chi_k \neq 0$. Hence Ψ is nonconstant. Choosing two fractional saddle phases where it takes different values produces two sequences along which the ratio to an uncorrected main term has different limits. This is a structural obstruction, not a matter of choosing a better constant.

20.5 A visually good quotient of exponentials has the wrong boundary scale

A proposed elementary approximation to the displaced bump has the form

$$Q(x) = \frac{1 + \exp(-|x|^{-\alpha})}{1 + \exp((1 - 2|x|)/(x^2 - |x|))}, \quad \alpha = \frac{3}{4}\sqrt{2\pi}. \quad (268)$$

It can look strikingly accurate on a plot. Put $y = x + 1$, the distance from the left support endpoint. For $0 < y \leq 1/4$ one has the rigorous bound

$$Q(-1 + y) \leq 2 \exp\left(-\frac{1}{2y}\right). \quad (269)$$

But [corollary 16.2](#) shows

$$\exp(-c/y) = o(F(y)) \quad (y \downarrow 0) \quad (270)$$

for every $c > 0$. Therefore

$$Q(-1 + y) = o(F(y)), \quad (271)$$

so the quotient is not asymptotically equivalent at the endpoint. This is a useful warning: a graph on a linear scale is almost blind to the distinction between $\exp[-\Theta((\log(1/y))^2)]$ and $\exp[-\Theta(1/y)]$ once both values are tiny.

20.6 What the formalization proves, and what it deliberately does not claim

The public aggregate contains exact proofs of the two source papers' named results, together with the additional dyadic, q -binomial, Legendre, and sharp-asymptotic theorems described here. It does not promote a source conjecture to a theorem, does not infer derivative bounds for an unspecified asymptotic remainder, and does not identify two global functions merely because they coincide on $[0, 1]$. These negative design decisions are part of the mathematics: they preserve the hypotheses needed by every later theorem.

21 Conclusion

The Fabius function is unusual not because it has one exotic property, but because several mathematical structures coincide in one object.

- A contraction on symmetric distribution functions constructs it and proves uniqueness.
- An infinite sum of independent uniform variables explains positivity, symmetry, and smoothing.
- Rvachev's fold is a compactly supported C^∞ density satisfying an exact functional-differential equation.
- Its Fourier transform is an entire sinc product with superalgebraic real-axis decay.
- Thue–Morse splitting creates a signed global solution and exact formulas for every derivative.
- Rational moment recurrences evaluate every dyadic argument exactly.
- A Taylor recursion turns the closed Thue–Morse formula into an efficient bit-sensitive algorithm.
- A negative-Laplace product and a lower-Lambert saddle expose the true endpoint scale, its Gamma–zeta periodic fluctuation, and its complete asymptotic expansion.
- A uniformly convergent Legendre series converts the same exact dyadic data into globally optimal polynomial approximants on the support.

The dyadic and asymptotic theories are not separate curiosities. The exact values $F(2^{-n})$ determine the moments and the Legendre coefficients; their logarithmic growth is governed by the same saddle geometry that controls arbitrary small arguments. The periodic term is the analytic shadow of dyadic self-similarity, while the Thue–Morse signs encode the same binary splitting algebraically. Seen together, these descriptions turn a famous “pathological example” into a remarkably coherent piece of analysis.

A Exact data

The following tables collect enough exact values to reproduce the examples in the text and to seed independent implementations.

A.1 Moment and half-moment sequences

n	c_n	d_n
0	1	1
1	$\frac{1}{9}$	$\frac{1}{2}$
2	$\frac{19}{675}$	$\frac{2}{5}$
3	$\frac{583}{59535}$	$\frac{1}{6}$
4	$\frac{132809}{32531625}$	$\frac{143}{6}$
5	$\frac{46840699}{24405225075}$	$\frac{1350}{19}$
		$\frac{270}{19}$

n	c_n	d_n
6	4068990560161	1153
	4133856862760625	23814
7	—	583
		17010

The c_n column satisfies (63); the d_n column satisfies (72). Substitution in (78) gives the next table.

A.2 Reciprocal powers and normalized odd integers

n	$F(2^{-n})$	$v_2(F(2^{-n}))$	R_n
0	1	0	—
1	$\frac{1}{2}$	−1	1
	$\frac{5}{2}$		
2	$\frac{72}{1}$	−3	5
	$\frac{1}{288}$		
3	$\frac{143}{2073600}$	−5	15
	$\frac{1}{19}$		
4	$\frac{33177600}{1153}$	−10	1001
	$\frac{1}{19}$		
5	$\frac{561842749440}{583}$	−15	5985
	$\frac{1}{19}$		
6	$\frac{179789679820800}{583}$	−22	2853675
	$\frac{1}{19}$		
7	$\frac{179789679820800}{583}$	−29	26261235
	$\frac{1}{19}$		

The valuation column is computed from (91); it is not obtained by factoring the displayed decimal-sized denominators. This is exactly the kind of redundancy that makes the table useful as a test vector.

B Proof dependency and Lean module guide

The Lean development is intentionally split into small files, while this article groups results by mathematical theme. The following map gives the principal entry points. File names are relative to

Analysis/FabiusFunction/Lean/FabiusFunction/.

Mathematical layer	Principal module(s)	Role in the proof
Exact sequences	<code>Arithmetic.lean</code>	Defines binary weight, Thue–Morse signs, c_n, d_n , division-free numerators, reciprocal values, closed dyadic formulas, and executable evaluators over $\mathbb{Q}$.
Analytic interface	<code>Basic.lean</code>	Separates bounded F , compactly supported up, and signed $\mathcal{F}$; defines Fourier products and moment integrals.

Mathematical layer	Principal module(s)	Role in the proof
Existence and uniqueness	<code>Existence.lean</code>	Constructs F as the fixed point of a strict contraction and proves uniqueness from dense dyadic values.
Differential equations	<code>Differential.lean</code>	Proves evenness and the global Rvachev equation, including the gluing argument at the fold point, then bootstraps to C^∞ .
Original characterization	<code>OriginalCharacterization.lean</code> , <code>OriginalUniqueness.lean</code>	Formalizes the compact-support characterization and proves uniqueness through the Fourier refinement product.
Probability	<code>ProbabilityRepresentation.lean</code>	Builds the infinite product probability space, proves independence, identifies the random series CDF with F , and obtains u as its density.
Centered splines	<code>FabiusUniformSpline.lean</code> , <code>FabiusComputability.lean</code>	Identifies the finite centered spline with a partial-sum CDF, proves the sandwich bound, and derives the explicit uniform error 2^{-p} .
Computable evaluator	<code>FabiusComputableSpline.lean</code>	Implements the signed-natural matched-grid spline by primitive recursion, certifies its rounding error, and proves sequential computability of the bounded Fabius function.
Finite convolutions	<code>ConvolutionApproximation.lean</code> , <code>StepApproximant.lean</code>	Connects polynomial densities, repeated convolutions, weak convergence, and the step approximants of the original paper.
Fourier and Poisson	<code>Fourier.lean</code> , <code>Poisson.lean</code>	Proves the sinc product, inversion, rapid decay, Poisson summation, cosine series, and partitions of unity.
Signed extension	<code>GlobalExtension.lean</code>	Proves local finiteness of the Thue–Morse translate sum, the global differential equation, and exact iterated-derivative formulas.
Moment arithmetic	<code>MomentPowerSeries.lean</code> , <code>HalfMomentGenerating.lean</code> , <code>Paper06487.lean</code>	Bridges the exact recurrences to integrals and entire generating functions and packages the arithmetic paper’s named statements.
Closed dyadic form	<code>DyadicCorrectness.lean</code> , <code>DyadicClosedForm.lean</code>	Proves affine Thue–Morse cancellation, the Taylor-block identity, and agreement of the fast evaluator with the closed formula.
Global dyadic values	<code>GlobalDyadic.lean</code>	Extends exact computation beyond the unit interval using triangular reduction and Thue–Morse signs.
q -binomial formulas	<code>FabiusQBinomialTaylor.lean</code> , <code>FabiusDyadicQBinomialScalar.lean</code>	Proves the finite q -binomial forms, arbitrary translation invariance, scalar extension, and dyadic-representation invariance.
Negative-Laplace product	<code>NegativeLaplace.lean</code> , <code>PeriodicCorrection.lean</code>	Splits the logarithmic product into its quadratic drift, periodic finite part, and exponentially small tail.

Mathematical layer	Principal module(s)	Role in the proof
Fourier analysis of Ψ	<code>PeriodicFourier.lean</code> , <code>PeriodicMean.lean</code> , <code>FabiusSharpConstant.lean</code>	Computes all Fourier coefficients and the mean of the periodic correction and identifies the sharp Euler–Stieltjes constant.
Lambert saddle	<code>FabiusLambertSaddle.lean</code> , <code>QuantitativeSaddle.lean</code> , <code>FabiusSharpAsymptotic.lean</code>	Introduces the lower-Lambert coordinate, performs exact Bromwich reduction, and proves the quantitative leading asymptotic.
Higher asymptotics	<code>FabiusFirstSaddleCorrection.lean</code> , <code>FabiusFullAsymptoticExpansion.lean</code> , <code>FabiusWikipediaExpansion.lean</code>	Derives the first periodic correction, every higher coefficient, the elementary log–log conversion, and the refutation of the uncorrected claimed error.
Decay comparisons	<code>FabiusDecayComparison.lean</code> , <code>FabiusQuotientExponentialMismatch.lean</code>	Places F between algebraic and ordinary flat-exponential scales and proves the endpoint failure of the visual quotient model.
Legendre expansion	<code>FabiusLegendreCoefficients.lean</code> , <code>FabiusLegendreSeries.lean</code>	Computes exact coefficients, proves endpoint-uniform absolute convergence, and establishes least-squares optimality.
Public aggregates	<code>Paper05442.lean</code> , <code>Paper06487.lean</code> , <code>PaperStatements.lean</code>	Expose source-facing theorem names and the paper-coverage interface from the public import <code>FabiusFunction</code> .

C Notation crosswalk

Different sources use F , φ , θ , and up for overlapping objects. The following crosswalk records the convention used here.

This article	Definition	Frequent source notation
$F(x)$	Bounded CDF on all of $\mathbb{R}$, constant 0 to the left and 1 to the right.	Fabius function on $[0, 1]$; sometimes the same letter is reused for the global extension.
$\text{up}(x)$	Even compactly supported density $F(1 - x)$ on $[-1, 1]$.	$\text{up}(x)$ or $\varphi(x)$.
$\mathcal{F}(x)$	Signed Thue–Morse extension, zero on $(-\infty, 0]$ and satisfying $\mathcal{F}' = 2\mathcal{F} \circ (2 \cdot)$.	$F(x)$ or $\theta(x)$ in papers that work globally.
c_n	Even moment $\int x^{2n} \text{up}(x) \, dx$.	c_n in the arithmetic paper.
d_n	Half moment $n \int_0^1 x^{n-1} \text{up}(x) \, dx$, with $d_0 = 1$.	d_n in the arithmetic paper.
Ψ	Centered or explicitly mean-separated one-periodic correction in the negative-Laplace finite part.	Often absent from elementary accounts.
λ	Exact positive solution of $\lambda 2^{-\lambda} = x$.	Lower-Lambert saddle coordinate $-W_{-1}(-x \log 2) / \log 2$.

When comparing formulas, one should first identify which global convention is in force and only then compare signs, supports, and endpoint values.

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